

A PROJECT REPORT

CAKE SHOP WEBSITE

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE

AWARD OF THE DEGREE OF

BACHELORE OF SCIENCE (INFORMATION TECHNOLOGY)

SUBMITTED BY

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SEAT NO:

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ABSTRACT

The demand for online shopping, particularly in the food and bakery industry, has increased significantly in recent years. With more customers seeking the convenience of ordering products online, cake shops can benefit from having a robust e-commerce platform to streamline their business operations. This Cake Shop Website is designed to provide a complete solution for customers to browse, customize, and purchase cakes with ease, while offering the business an efficient system for managing orders, inventory, and customer interactions.

This website enables customers to select cakes based on categories, personalize their orders by choosing flavors, sizes, and custom messages, and securely place their orders through an integrated payment gateway. The platform also provides real-time updates on the status of the order, ensuring a transparent and smooth user experience. The admin panel allows cake shop owners to efficiently track and manage orders, update inventory, and monitor business performance.

This paper analyzes the various features and functionalities that make this e-commerce solution a comprehensive tool for both the cake shop and its customers. By integrating modern web technologies such as MongoDB, Express.js, React, and Node.js (MERN stack), the platform ensures scalability, performance, and security. This project aims to address the growing need for digital solutions in the cake industry, enhancing both customer satisfaction and business efficiency.

ACKNOWLEDGEMENT

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Lastly, I am grateful to the online development communities, forums, and resources that provided useful technical insights, without which this project would not have been possible.

Thank you all for your contributions, encouragement, and belief in this endeavor.

DECLARATION

I here by declare that the project entitled, “CAKE SHOP WEBSITE” has not been in any case duplicated to submit to any other university of the award of any degree. To the best of my knowledge other than me, no one has submitted to any other University.

The project is done in partial fulfilment of the requirements for the award of degree of **BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)** to be submitted as final semester project as part of our curriculum.

Chinmay Abhijeet Koshe

Signature

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CHAPTER I

INTRODUCTION

A Cake Shop Website is an online platform designed to offer customers a seamless and interactive experience for browsing, customizing, and purchasing cakes from the comfort of their homes. These websites are developed with the primary goal of providing an easy-to-use digital solution for cake shops to expand their reach, while offering customers the convenience of placing orders for customized cakes for various occasions. By combining modern web technologies with an understanding of the business and customer needs, Cake Shop Website s aim to enhance the shopping experience and streamline operations for the business.

The platform is specialized to offer a comprehensive catalogue of cakes, along with customization options that allow customers to personalize their orders based on preferences such as Flavors, sizes, and designs. This type of website is particularly focused on addressing the growing demand for online shopping in the food industry, while providing cake shops with tools to manage orders and inventories efficiently. Customers can easily browse through the cake selection, place an order, and track status in real-time, ensuring a smooth and satisfying shopping experience.

The Cake Shop Website aims to create a flexible and scalable solution that caters to both customer preferences and business requirements. By integrating features such as product customization and an intuitive user interface, the platform empowers cake shops to improve their online presence, enhance customer satisfaction, and streamline their operations, ultimately contributing to the growth of their business.

1.1 Background

The rise of e-commerce has revolutionized how businesses interact with customers, particularly in the food industry. As consumer preferences shift towards online shopping for convenience and accessibility, the need for businesses to establish an online presence has never been more critical. The bakery and cake industry, known for its demand for personalized products such as custom cakes for special occasions, has also embraced this shift.

A traditional brick-and-mortar cake shop faces limitations in reaching a wider audience and streamlining its operations. Customers often face challenges like limited store hours, in-person communication issues, or miscommunications regarding customized orders. In response to these challenges, developing an online platform for a cake shop ensures that customers can browse, customize, and order cakes effortlessly while the shop manages inventory and order processing more efficiently.

The Cake Shop Website aims to address these challenges by providing a comprehensive digital solution that allows users to easily view available products, customize orders according to their preferences, and complete secure transactions online. By leveraging modern web technologies, this platform not only enhances the customer experience but also optimizes the shop's operational efficiency. The implementation of features such as reducing the time for selection, order tracking, keeping records, order placing allows the shop to expand its reach, reduce errors, and offer a seamless online shopping experience.

In summary, the development of a Cake Shop Website serves to bridge the gap between traditional in-store purchasing and the growing demand for online convenience, creating a more accessible and efficient customer experience.

1.2 Objectives

The Cake Shop Website is designed with a customer-centric approach, aiming to streamline the process of ordering cakes online while improving the operational efficiency of the business. It leverages modern technologies to create a seamless shopping experience for users and a robust management system for the cake shop. Below are the key objectives of the project:

1. Enhance User Experience:

- Provide customers with an intuitive and visually appealing interface to browse through a variety of cakes and options.
- Ensure ease of navigation and customization options, offering a user-friendly process for selecting cake designs, Flavors, sizes, and personalized inscriptions.
- A simple and enjoyable online shopping experience, leading to higher customer satisfaction and retention.

2. Enable Real-Time Order Management:

- Implement real-time updates for customers on the status of their orders, from confirmation to delivery.
- Keep customers notified.
- Minimize errors and delays in fulfilling orders through a real-time order management system for the business.

3. Facilitate Custom Cake Ordering:

- Allow users to personalize their cake orders with various options such as flavors, sizes, decorations, and custom messages.
- Simplify the customization process while offering diverse choices to meet individual preferences.

4. Improve Business Insights and Analytics:

- Implement tools for the business to generate sales reports, monitor popular products, and track customer preferences.
- Use analytics to provide insights that help the business make informed decisions, optimize offerings, and increase profitability.
- Allow shop owners to track business performance, improving overall efficiency and scalability.

5. Promote Brand Visibility and Reach:

- Use the website to enhance the online presence of the cake shop, attracting a broader audience beyond the local market.
- Integrate marketing tools such as SEO, social media sharing, and promotions to drive traffic and customer engagement.
- Increase the visibility of the brand, allowing the shop to expand its customer base.

6. Ensure Data Security and Privacy:

- Implement robust data security protocols to protect customer information, including personal data and order history.
- Adhere to privacy regulations and industry standards, ensuring user confidence in the security of their interactions with the platform.
- Prevent unauthorized access and protect sensitive information through encrypted communication and secure authentication methods.

7. Enable Scalability and Flexibility:

- Design the website architecture to accommodate future growth, including additional features, higher traffic, and expanded product offerings.
- Ensure the platform is scalable to support an increasing number of users and orders as the business expands.

By focusing on these objectives, the Cake Shop Website will provide a comprehensive solution that not only enhances customer satisfaction but also improves the efficiency and profitability of the cake shop, ensuring long-term success in the competitive e-commerce market.

1.3 Purpose, Scope, and Applicability

1.3.1 Purpose

The purpose of the Cake Shop Website is to create a comprehensive and user-friendly online platform that simplifies the process of cake ordering for customers while optimizing the operational efficiency of the cake shop. This website is designed to address the needs of both customers seeking customized cakes for special occasions and shop owners looking to streamline their order management and inventory control. The platform not only enhances the customer experience but also provides the shop with tools to expand its market reach and improve business performance.

By developing this website, the goal is to offer a seamless solution that integrates key e-commerce functionalities, such as product browsing, order customization, secure payment processing, and real-time tracking, while ensuring ease of use and accessibility for all users. The Cake Shop Website aims to provide a holistic digital solution that meets the demands of the modern consumer, ensuring convenience, personalization, and security in every aspect of the shopping experience.

1.3.2 Scope

The scope of the Cake Shop Website is extensive, encompassing both frontend and backend functionalities. On the frontend, the website will offer an intuitive and responsive interface that allows users to browse cakes, customize orders, and complete purchases with ease. This will be supported by features such as product filtering, personalization options, and an integrated shopping cart.

On the backend, the website will provide shop administrators with tools to manage orders, monitor inventory, and track sales performance. Real-time updates, order tracking, and a user-friendly admin dashboard will further enhance the operational capabilities of the business.

This project is designed to be scalable and adaptable, allowing the cake shop to expand its product offerings, support multiple locations, and manage higher volumes of orders as the business grows. The scope also includes potential future integrations, such as loyalty programs, delivery service partnerships, and advanced analytics for customer insights, secure payment gateways & improved navigation.

1.3.3 Applicability

The Cake Shop Website is applicable to a wide range of cake shops and bakeries, particularly those seeking to modernize their operations and expand their customer base through online sales. The platform is versatile and can be customized to suit the specific needs of small, medium, or large cake shops. It provides a solution that can be adapted to various business models, whether the focus is on local deliveries, in-store pickups, or shipping to distant locations.

This system can also be extended to other food-related businesses, such as pastry shops, dessert cafes, or specialty bakeries, making it a flexible and widely applicable solution in the food and beverage industry. The website's design ensures that it can be used across different devices, providing accessibility and convenience for users on mobile, tablet, or desktop.

CHAPTER 2

SURVEY OF TECHNOLOGIES

This project explores the MERN stack to develop a robust Cake Shop E-Commerce website, ensuring efficiency, scalability, and high performance.

2.1 Overview of MERN Stack

The MERN stack, comprising MongoDB, Express.js, React, and Node.js, is a popular choice for full-stack web applications:

- **MongoDB:** A NoSQL database using BSON for flexible data storage, ideal for managing unstructured data like order details, reviews data and customer profiles.
- **Express.js:** A lightweight framework for Node.js that simplifies API creation and server-side operations, facilitating communication between back-end and front-end.
- **React:** A front-end library for building dynamic user interfaces, allowing for reusable components that enhance user interactions.
- **Node.js:** A server-side runtime environment that efficiently handles concurrent requests, managing API requests and real-time updates.

2.2 Justification for Using the MERN Stack

The MERN stack is chosen for several reasons:

- **Unified Language:** Full JavaScript development simplifies application management and reduces complexity.
- **Scalability:** MongoDB's structure supports diverse data types and accommodates business growth.
- **Performance:** The event-driven architecture of Node.js and Express.js provides high performance and real-time capabilities.
- **Component Reusability:** React's architecture allows for efficient UI updates, contributing to a responsive experience.
- **Community Support:** The active ecosystem surrounding the MERN stack accelerates development and feature integration.

2.3 Comparison with Other Technology Stacks

- **LAMP Stack:** Traditional and reliable, but less scalable and performant than the MERN stack.
- **MEAN Stack:** Similar to MERN but uses Angular, which has a steeper learning curve and less flexibility than React.
- **Django Stack:** Rapid development with Python, but its separation from JavaScript complicates full-stack development compared to MERN.

2.4 Key Technologies Used for Additional Features

To enhance functionality, the following technologies will be integrated:

- **Google Maps:** Allows customers to locate the store and set delivery addresses.
- **User Authentication:** JWT for secure login and session management.
- **Cloud Hosting:** Deployment on Vercel to ensure scalability and uptime.

CHAPTER 3

REQUIREMENTS AND ANALYSIS

This chapter outlines the detailed analysis of functional and non-functional requirements for the Cake Shop Website. Key features include user account management, cake customization options, a product catalogue, and secure checkout. Emphasis is placed on user-friendliness and seamless navigation, integrating Google Maps for location accuracy and an admin panel for efficient order and inventory management. The analysis leverages the Waterfall Model to ensure sequential and structured development.

Analysis Model: Waterfall Model

This model allows a structured approach, ensuring each phase is completed before moving on to the next, supporting a clear and organized development process for the project.

This **Waterfall Model** provides with a sequential software development process with distinct phases:

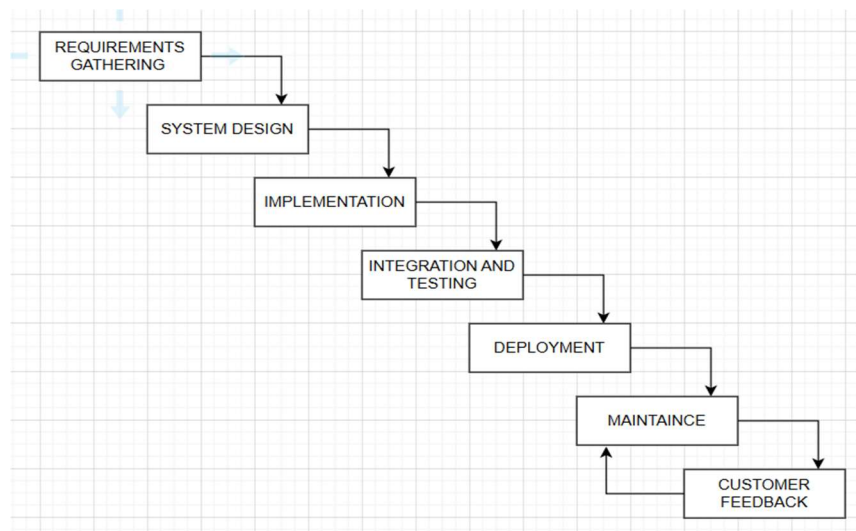


FIG NO. 1: WATERFALL MODEL

The waterfall model is a sequential design process, often used in software development processes, in which progress is seen as flowing steadily downwards (like a waterfall) through the phases of Conception, Initiation, Analysis, Design, Construction, Testing and Maintenance

Advantages of the Waterfall Model:

Simple and easy to understand and use.

Easy to manage due to the rigidity of the model each phase has specific deliverables and a review process.

Phases are processed and completed one at a time.

Works well for smaller projects where requirements are very well understood.

FEASIBILITY STUDY

A feasibility study evaluates the practicality and potential success of the Cake Shop Website project. The study focuses on different areas to ensure that the project meets all requirements efficiently and sustainably.

ECONOMICAL FEASIBILITY

Assesses the cost-effectiveness of developing the website and if the investment aligns with the expected benefits.

Considers initial setup costs, hosting fees, maintenance, and potential long-term savings or benefits.

Helps to evaluate the return on investment (ROI) and financial viability.

TECHNICAL FEASIBILITY

Determines if the project can be completed with the current technology stack (MERN) and resources.

Evaluates infrastructure needs such as servers and assesses if team skills align with the development requirements.

Ensures that the chosen technologies can support the website's goals and performance

SOCIAL FEASIBILITY

Analyses the potential acceptance and user experience of the website.

Considers user expectations regarding accessibility, ease of use, and whether the website serves customer needs effectively.

Ensures that the project has a positive impact on users and meets community standards

3.1 Problem Definition

The Cake Shop Website is designed to modernize and simplify the cake ordering experience for customers. Traditional in-person or phone-based ordering can be inconvenient, time-consuming, and limits customer engagement with the shop's offerings. This website allows customers to explore a diverse product catalogue and pricing, and customize orders based on their preferences. It also provides essential shop information such as location, hours, and contact details. By streamlining the order placement process, enhancing customer accessibility, and improving shop management of orders, the website serves as an efficient solution that elevates both user convenience and business operations.

The Cake Shop Website also incorporates a user-friendly interface to ensure that customers of all ages and technical backgrounds can navigate it easily. Additionally, the website enhances the brand's visibility, attracting a wider audience through an online presence. By integrating features such as user authentication and order placing, the platform offers a secure, personalized experience. These elements combined make the website a valuable tool for boosting customer satisfaction and operational efficiency.

3.2 Requirement Specification

The Cake Shop Website requires a comprehensive, user-friendly platform where customers can view a catalogue of cakes, customize their orders, and complete purchases. It should enable customers to manage accounts, place order, order personalized cakes, track order status, and access shop details, including location via Google Maps. Additionally, the website should feature an admin panel for managing orders, placing order from store, and product details, providing efficient backend operations for the cake shop staff. The system must ensure accessibility, reliability, and seamless functionality for both customers and the admin team.

Functional Requirements:

- **User Registration & Authentication:** Secure login and account creation with email verification.
- **Product Catalogue:** A visually organized catalogue with filtering options.
- **Cake Customization:** Options for size, flavor, and custom messages.
- **Shopping Cart & Checkout:** Easy order finalization and payment integration.
- **Admin Management:** Order management & placing from store through an admin panel.
- **Google Maps Integration:** Assistance with locating the shop or setting delivery addresses.

Non-Functional Requirements:

- **Performance:** Fast loading under high user traffic.
- **Scalability:** Handle increased orders and product listings.
- **Security:** Protect user data with SSL, encryption, and secure authentication.
- **Reliability:** Consistent uptime and prompt recovery from failures.

Existing System with Limitations

The current system for ordering cakes involves in-person visits or phone calls, which is time-consuming and limits customer convenience. Customers cannot view the shop's full range of products or customize cakes easily without visiting. Order tracking and status updates are not readily available, leading to potential confusion. Additionally, manual order handling may lead to errors and makes order management inefficient for the shop staff.

Proposed System

The proposed Cake Shop Website addresses these limitations by offering an online platform for browsing products, placing and customizing orders. Customers can create accounts, save preferences, and receive real-time updates on their orders. The system includes a secure checkout process and integrates Google Maps to show the shop's location. An admin panel enables staff to manage orders, update inventory, and view sales, streamlining shop operations, managing in store orders and enhancing the customer experience.

3.3 Planning and Scheduling

Planning:

The planning phase for the Cake Shop Website involves defining the project's scope, identifying the necessary resources, and establishing a timeline for completion. Key components of the planning process include:

Project Scope: Clearly outline the features and functionalities, including user management, product catalogue, order customization, and admin panel.

Risk Assessment: Evaluate potential risks (e.g., technical challenges, timeline delays) and develop mitigation strategies.

Scheduling:

The scheduling phase involves creating a timeline for the project, detailing when each phase of development will occur. This ensures that the project stays on track and deadlines are met. Key components of the scheduling phase include:

- **Task Breakdown:** Divide the project into smaller tasks with specific deliverables.
- **Timeline Establishment:** Set estimated start and end dates for each task.
- **Resource Assignment:** Assign team members to specific tasks based on their expertise.

Gantt chart:

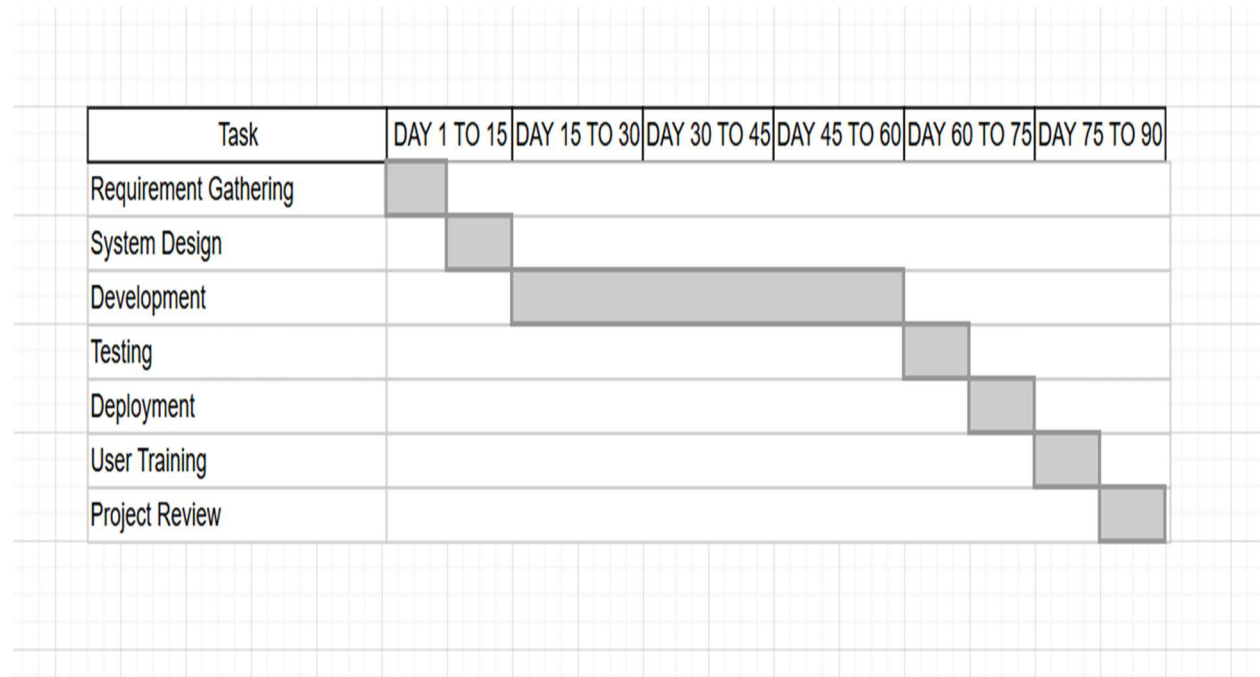


Fig No: 2. Gantt chart

3.4 Software and Hardware Requirements

3.4.1 Software Requirements

Operating System	:	Windows/ macOS
User interface	:	HTML, CSS, and JavaScript frameworks
Database	:	MongoDB
Documentation Tool	:	MS Word
Server	:	Node.js with Express.js

3.4.2 Hardware Requirements

Processor	:	Intel i5 or AMD Ryzen 5
RAM	:	8 GB (minimum)
Hard Disk	:	256 GB SSD (minimum)
Monitor	:	Standard Monitor
Keyboard	:	Standard ergonomic keyboard
Mouse	:	Optical mouse

3.5 Preliminary Product Description

The **Cake Shop Website** is a web-based application designed to facilitate the online ordering and customization of cakes. This platform enhances the shopping experience by allowing customers to browse a wide selection of cakes, personalize their orders, and complete transactions conveniently from their devices.

The application leverages **MongoDB** as its database management system, enabling flexible data storage and quick access to information such as user profiles, product catalogues, and order histories. The website features a user-friendly interface, making navigation intuitive for users of all ages and technical backgrounds.

Project Category:

This project is based on a NoSQL database system, specifically **MongoDB**, which is used for all transaction statements. MongoDB is an open-source document-oriented database designed for scalability and flexibility, allowing for the storage of data in JSON-like format.

Services provided:

User Account Management: Users can create accounts, log in, and manage their profiles, which include personal details and order history.

Product Catalogue: A dynamic catalogue displaying a variety of cakes with descriptions, images, and prices, allowing users to easily browse options.

Order Customization: Customers can personalize their cake orders based on preferences such as flavour, size, and decoration, ensuring tailored experiences.

Shopping Cart Functionality: Users can add items to their shopping cart, view selected products, modify quantities, and remove items before checkout.

Easy Checkout Process: A streamlined and secure checkout process that order confirmation via provided data input fields.

Order Tracking: Customers can track the status of their orders in real time, receiving updates on approval and completion.

Admin Panel: A comprehensive interface for shop staff to manage inventory, process orders, and manage in-store orders.

Google Maps Integration: Integration with Google Maps to show the shop's location and provide navigation assistance for customers.

3.6 Conceptual Models

The Cake Shop Website features a three-tier architecture: a presentation layer for user interaction, an application layer for processing orders and managing data, and a data layer using MongoDB for storage. Key functions include user account management, order customization, and an admin panel for order management.

3.6.1. Use Case Diagram

A use case diagram is a diagram that shows a set of use cases and actors and relationships.

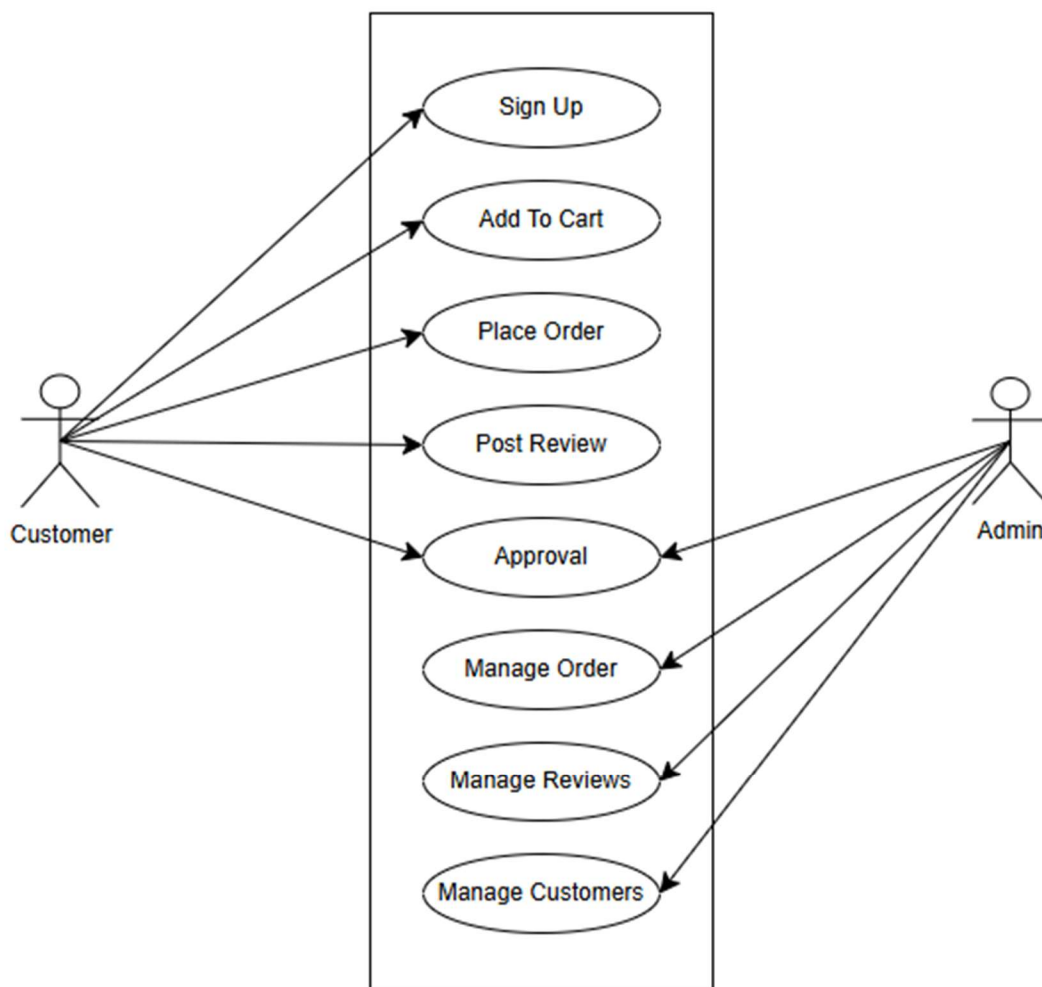


Fig No: 3. Use Case Diagram

3.6.2 Activity Diagram

- An activity diagram shows the flow from activity to activity. An activity is an ongoing non- atomic execution within a state machine.
- Activities ultimately result in some action, which is made up of executable atomic computations that result in a change in state of the system or the return of a value.

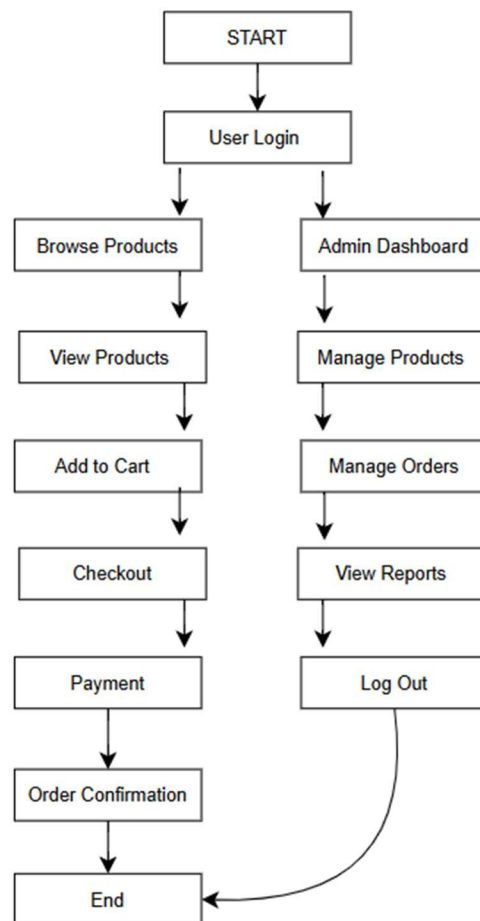


Fig No: 4. Activity Diagram

3.6.3 Data Flow Diagram

- A data flow diagram (DFD) is a graphical representation of the “flow” of data through an information system.
- DFDs can also be used for the visualization of data processing.

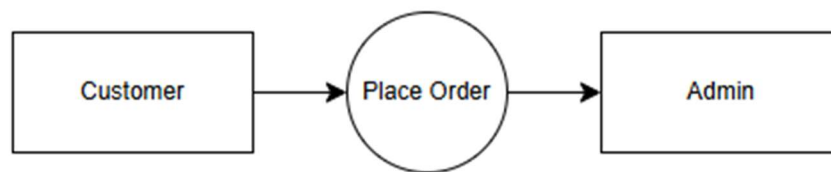


Fig No: 6. Zero Level Data Flow Diagram

3.6.4 Flowchart

3.6.4.1 Server-Side Flowchart

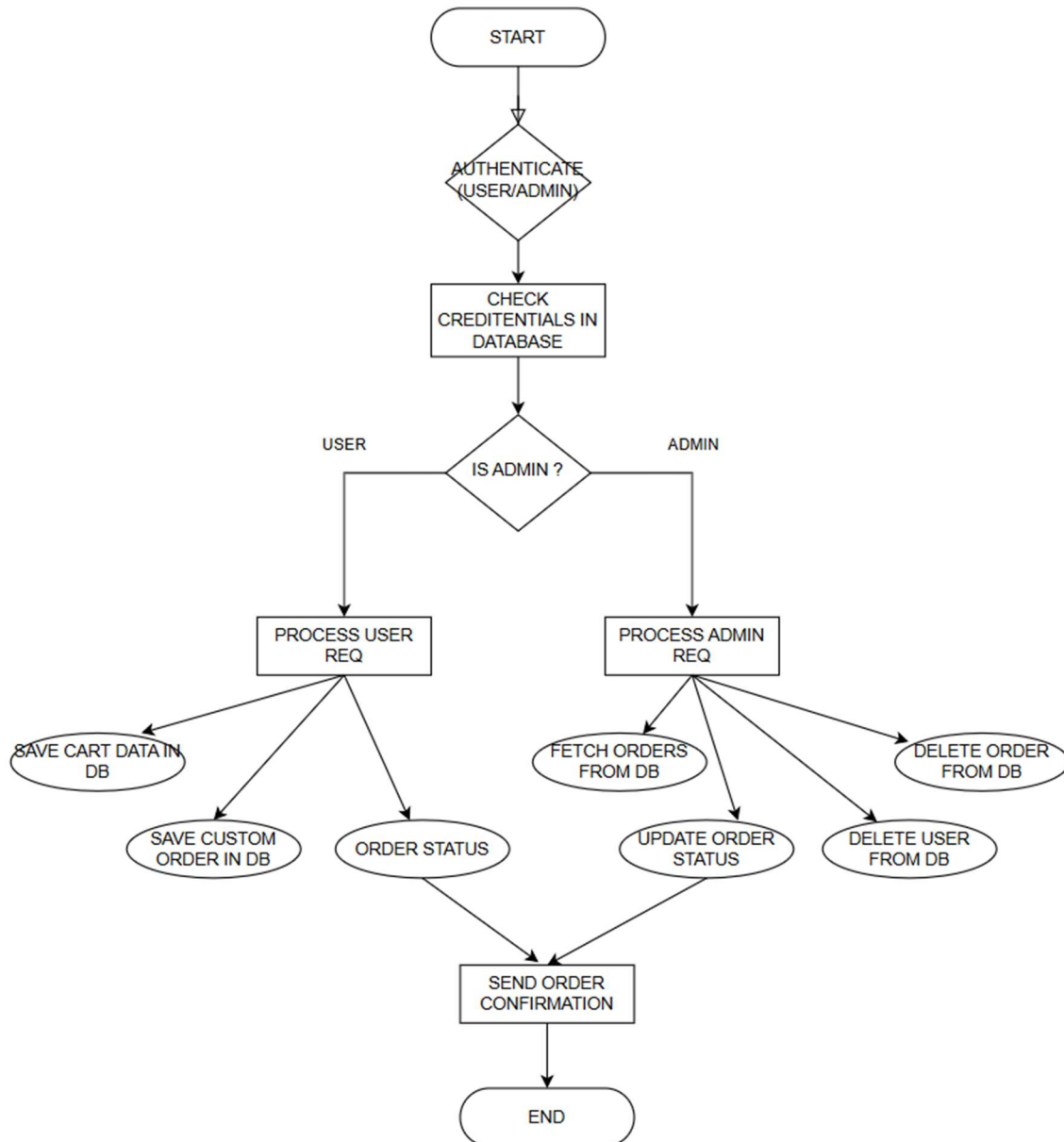


Fig No: 7. Server-Side Flowchart

3.6.4.2 Client-Side Flowchart

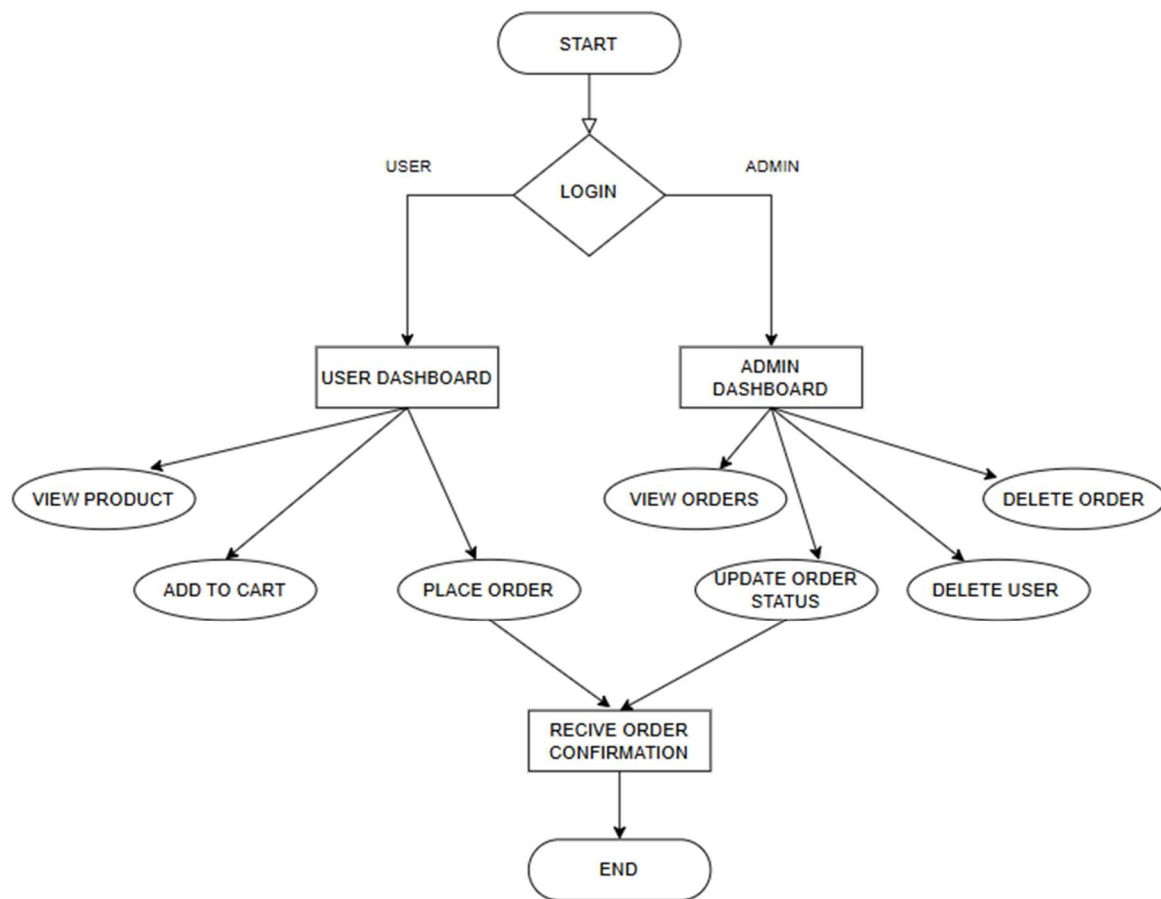


Fig No: 8. Client-Side Flowchart

3.6.5 Class Diagram

A class diagram represents the structure of the system. It shows set of classes, interfaces, and relationships between them.

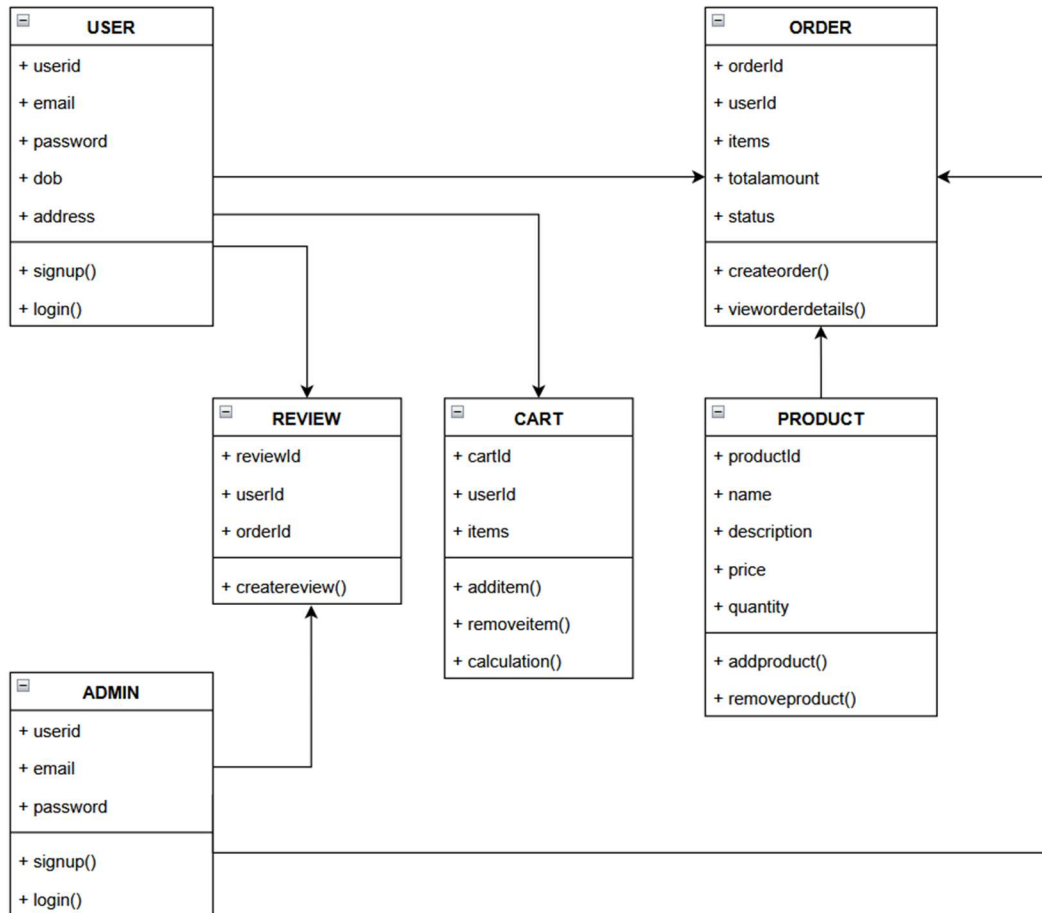


Fig No: 9. Class Diagram

3.6.6 Sequence Diagram

An interaction diagram shows an interaction, consisting of a set of objects and their relationships, including the messages that may be dispatched among them. A sequence diagram is an interaction diagram that emphasizes the time ordering of messages. Graphically, a sequence diagram is a table that shows objects arranged along x-axis and messages, ordered in increasing time, along the y-axis.

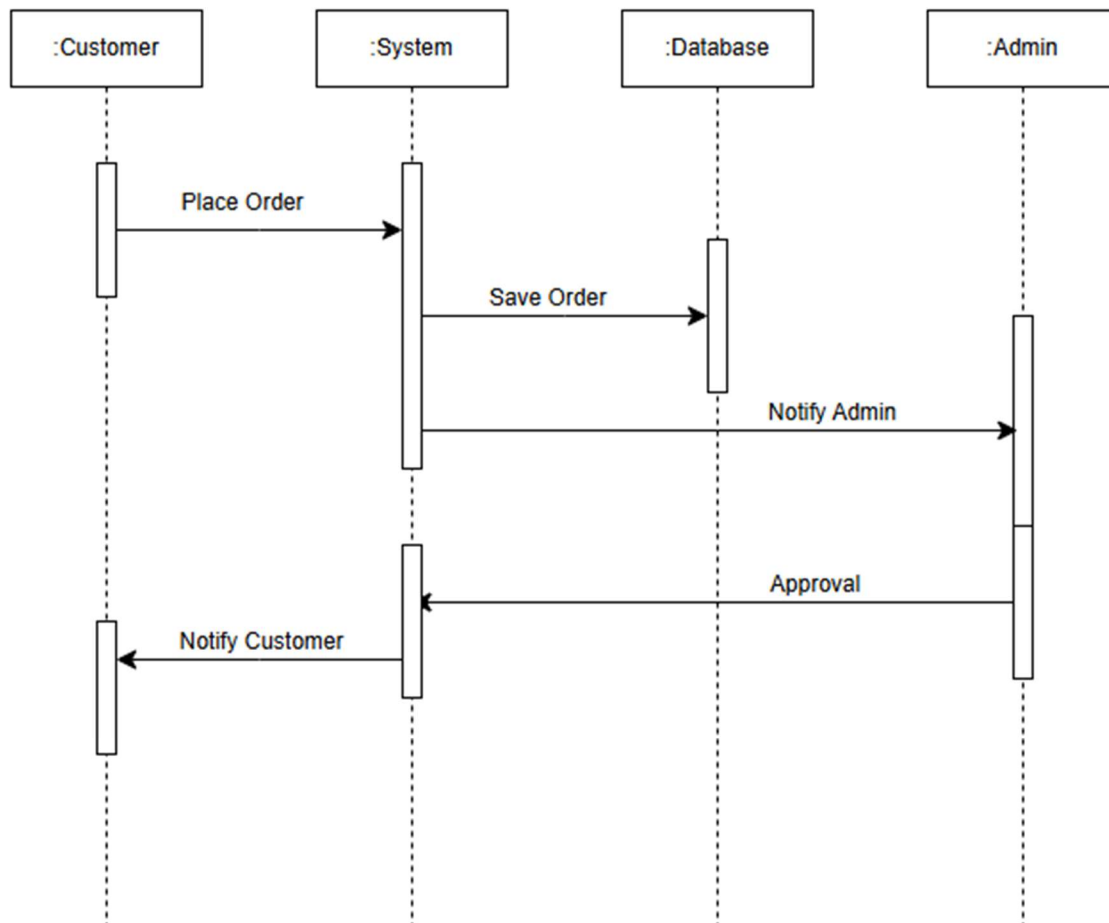


Fig No: 10.Sequence Diagram

3.6.7 Entity-Relationship (E-R) Diagram

- The ER Diagram is structured design of database. It acts as a framework created with specialized symbols for purpose defining relationship between The database entities. ER diagram is based on three principal components: entities attributes and relationships

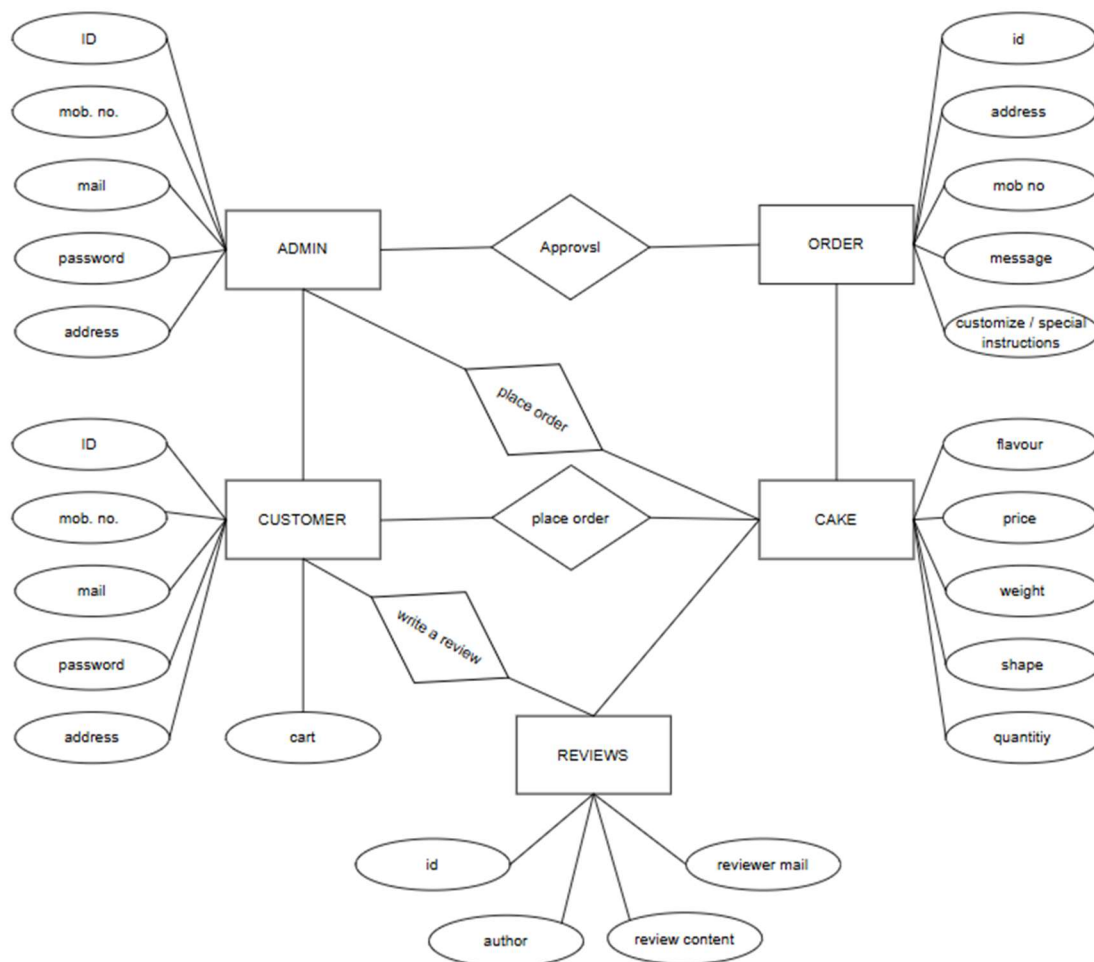


Fig No: 11. E-R Diagram

CHAPTER 4

SYSTEM DESIGN

4.1 Basic Modules

The Cake Shop Website consists of several modules that collaborate to deliver a seamless user experience. Each module handles specific functionalities, ensuring system efficiency and user satisfaction.

4.1.1 Admin Module

The admin module provides tools for managing the website's operations effectively.

Features:

1. User Management:
 - View or delete user accounts.
2. Product Management:
 - View cakes, their order ID, and images.
 - View customization options.
3. Order Management:
 - View all orders placed by users.
 - Update order statuses (e.g., pending, shipped, completed).
 - Delete misplaced orders.
4. Inventory Management:
 - Monitor real-time stock levels of products.
 - Receive alerts when inventory falls below a certain threshold.
 - Manage supplier details for restocking.
5. Reporting and Analytics:
 - Export reports in CSV or PDF formats for further analysis.

4.1.2 User Module

The user module is designed to deliver an intuitive and personalized shopping experience for customers.

Features:

1. Account Registration and Login:
 - Enable users to create accounts with email verification.
 - Allow secure login using hashed credentials (e.g., bcrypt).
 - Offer a view-only option for guest users.
2. Profile Management:
 - Allow users to update personal information (e.g., name, address).
 - View their order history with details like order date, status, etc.
3. Product Browsing:
 - Display a catalog of cakes with images, descriptions, and prices.
 - Implement search functionality with filters (e.g., by flavor, size, or price).
4. Order Customization:
 - Provide options for users to customize cakes (e.g., flavors, toppings, designs).
5. Shopping Cart:
 - Allow users to add, edit, or remove items from the cart.
 - Calculate total prices dynamically.
6. Order Tracking:
 - Provide real-time updates on order statuses (e.g., processing, shipped, delivered).
 - Notify users via email about order status changes.
 -

4.1.3 Reporting Module

The reporting module provides insights for admins to make informed decisions.

Features:

- Sales Reports:
 - Show sales trends for the most ordered cakes (daily or all-time).

4.2 Data Design

The data design defines how information is organized, stored, and accessed using MongoDB.

4.2.1 Schema Design

The database includes the following key collections:

1. Users Collection:

Field Name	Type	Required	Notes
name	String	Yes	First name of the user.
surname	String	Yes	Last name of the user.
email	String	Yes	Unique email address, validated with regex, automatically converted to lowercase.
mobile	String	Yes	10-digit mobile number, validated with regex.
dob	Date	No	Date of birth of the user.
address	String	Yes	Address of the user.
password	String	Yes	Password (usually hashed before storage).
cartProducts	Array of Objects	No	List of products in the user's cart.
orderID	String	Yes	Unique order ID for the product in the cart.
name	String	No	Name of the product in the cart.
quantity	Number	No (Default: 1)	Quantity of the product in the cart.
weight	String	No	Weight of the product in the cart.
shape	String	No	Shape of the product in the cart.
img	String	No	Image URL of the product in the cart.
price	Number	No	Price of the product in the cart.
Role	String	No (Default: user)	User's role, either user or admin.

2. Orders Collection:

2.1. Order Schema:

Field Name	Type	Required	Notes
userEmail	String	Yes	Indexed and automatically converted to lowercase using a setter.
orderItems	Array of Objects	Yes	Contains details about the ordered items.
orderId	String	Yes	Unique identifier for each item in the order.
name	String	Yes	Name of the ordered cake/item.
shape	String	Yes	Shape of the cake (e.g., round, square).
quantity	Number	No (Default: 1)	Quantity of the item ordered.
price	Number	Yes	Price per item.
weight	String	Yes	Weight of the cake/item.
image	String	Yes	URL or path for the item's image/design.
totalAmount	Number	Yes	Total price of the order.
deliveryAddress	String	Yes	Address for delivery.
paymentMethod	String	Yes	Payment mode: COD (Cash on Delivery) or Online.
status	String	No (Default: Pending)	Order status: Pending, Completed, Cancelled, Accepted.
createdAt	Date	No (Default: Date.now)	Timestamp for when the order was created.
cakeMessage	String	Yes	Custom message for the cake, maximum length of 100 characters.
orderDate	Date	Yes	Date when the order was placed.
orderTime	String	Yes	Time when the order was placed.

2.1. Customized Order Schema:

Field Name	Type	Required	Notes
name	String	Yes	Name of the user/customer.
email	String	Yes	Automatically converted to lowercase using a setter.
phone	String	Yes	Optional phone number.
address	String	Yes	Delivery address.
size	String	Yes	Cake size (e.g., small, medium, large).
cakeType	String	Yes	Type of cake (e.g., birthday, wedding).
flavor	String	Yes	Flavor of the cake (e.g., vanilla, chocolate).
message	String	No	Custom message for the cake.
specialInstructions	String	No	Any additional instructions for customization or delivery.
deliveryDate	Date	Yes	Required date for the delivery.
approvalStatus	String	No (Default: pending)	Tracks order approval status (e.g., pending, approved, rejected).
price	Number	No (Default: 0)	Total price of the order.
imageOrDesign	String	No	Optional link or name of the design/image for the cake.

3. Review Collection:

Field Name	Type	Required	Notes
orderId	ObjectId	Yes	Reference to the Order model, indicates the associated order.
content	String	Yes	The review content provided by the user.
authorName	String	No	Name of the author (optional, from the authenticated user).
authorEmail	String	No	Email of the author (optional, automatically converted to lowercase).
createdAt	Date	No (Default: Date.now)	Timestamp for when the review was created.

Relationships:

- One-to-many relationship between Users and Orders: A user can place multiple orders.
- Many-to-many relationship between Orders and Products: Each order contains one or more products, and a product can be part of multiple orders.

4.2.2 Algorithms Design

Key algo In cake shop website include:

User Authentication:

- Validates login with bcrypt (hashed passwords) and manages sessions with JWT.
- Implements security measures like login error messages and account logout.

Order Processing:

- Calculates the total price for an order, considering items and customization.
- Validates and updates the order collection.
- Updates order status (e.g., Pending, Accepted and Cancelled) and notifies users.

Product Search:

- Performs full-text and filtered searches based on attributes like description and flavor.
- Supports sorting, pagination, and autocomplete for efficient user experience.

Cart Management:

- Handles adding, updating, and removing items from the cart.
- Dynamically updates the total price based on cart changes, with persistence across sessions or user logins.

Order Tracking:

- Tracks and updates order status (e.g., approval).
- Sends notifications to users about status changes.

Review and Rating:

- Allows users to submit reviews for cakes.

Forgot Password:

- Allows users to reset their password securely via email with a time-limited token.

4.3 User Interface Design

The User Interface (UI) Design for the Cake Shop Website focuses on creating an intuitive and visually appealing experience for both admins and users. The design elements are structured to enhance usability and accessibility.

Admin UI Design

Frontend Structure:

- **Dashboard:** The admin interface begins with a comprehensive dashboard that aggregates key information, such as recent orders, sales statistics, and inventory levels. This serves as the main entry point for admins to navigate through various management sections.
- **Responsive Design:** Just like the user interface, the admin dashboard is also built with a mobile-first design using Tailwind CSS. This ensures that admins can efficiently manage the system even on smaller devices, though it's optimized for desktop use.
- **Components:** Admin UI components are structured to display detailed data in a user-friendly manner, such as:
 - Tables for managing users, orders, and inventory.
 - Forms for adding or updating product details, user information, and order status.
 - Charts/Reports to visualize sales data, order trends, and inventory statistics.

Admin Interaction:

- **User Management:** Admins can view, edit, or delete user accounts. This allows them to manage customer profiles, ensure data accuracy, and handle any necessary customer support tasks.
- **Product Management:** Admins can add, edit, or remove products from the catalog, managing their names, descriptions, images, prices, and stock levels.
- **Order Management:** Admins can view all customer orders, update their statuses (e.g., Pending, Shipped, Completed), and handle any issues like canceled or misplaced orders.
- **Inventory Management:** Admins are alerted when stock levels fall below a certain threshold, and they can manage supplier details for restocking.

- **Reporting:** The admin can generate detailed reports on sales performance, product trends, and customer data. These reports can be exported in CSV or PDF formats for further analysis.

The admin UI focuses on making the management process efficient, with intuitive navigation and clear information display to assist admins in their tasks.

User UI Design

Frontend Structure:

- **Home Page:** The landing page showcases featured products, popular cakes, and special offers. It acts as the entry point for users to explore the website and navigate to different sections.
- **Responsive Design:** The user interface is built with a mobile-first approach using Tailwind CSS, ensuring that the website provides a smooth and visually appealing experience on all devices, from mobile phones to desktops.
- **Components:** The user interface is composed of reusable components like:
 - Product Cards for displaying cakes with images, descriptions, and prices.
 - Buttons and Forms for actions like adding items to the cart, customizing cakes, and checking out.
 - Navigation Menus for easy access to product categories, cart, and account management.

User Interaction:

- **Product Browsing:** Users can view the product catalog, which includes cakes with images, detailed descriptions, and price information. They can filter products by category, flavor, or price.
- **Product Detail Page:** Each product has a dedicated page that provides more details, customization options, and a way to add the product to the cart.
- **Account Registration and Login:** New users can register by providing their details and verifying their email. Existing users can log in securely using hashed passwords (with bcrypt). Users can also view products without signup.
- **Shopping Cart:** Users can view items added to their cart, update quantities, remove items, and see the dynamically calculated total price. The cart is persistent across sessions, so users can log in and find their cart intact.
- **Order Customization:** Users can customize cakes by selecting different flavors, sizes, toppings, and designs before adding them to the cart.

- **Order Tracking:** Once an order is placed, users can track its status (e.g., Pending, accepted, cancelled) and receive notifications via email regarding status changes.
- **Profile Management:** Users can view and update their personal information, such as name, address, and contact details. They can also check their order history.

The user interface is designed to be intuitive, ensuring that customers can easily browse products, customize their orders, and manage their profiles with minimal effort.

4.4 Security Issues

1. **Authentication and Session Management:** Use JWT tokens for secure user authentication and session management, ensuring token expiration and secure transmission over HTTPS.
2. **Data Protection:** Safeguard sensitive data with password hashing (e.g., bcrypt) and encryption (e.g., AES) for both storage and transmission to prevent unauthorized access.
3. **Role-Based Access Control (RBAC):** Implement RBAC to restrict access to sensitive resources based on user roles, following the least privilege principle to minimize unauthorized access.

4.5 Test Cases Design

Test cases ensure the application functions correctly under various scenarios:

Test Case ID	Test Case Title	Description	Expected Result
TC001	Navbar Rendering on Home Page	Verify if the Navbar is rendered correctly on the Home page.	Navbar should be properly rendered with all expected elements visible.
TC002	Hero and Featured Sections	Check if Hero, Best Sellers, Designs, Reviews, Process, Call-to-Action, and Footer are rendered.	All sections should be displayed correctly and fully functional.
TC003	Carousel Functionality	Validate if the carousel on the Design page is displayed and clickable.	Carousel should render properly and allow navigation between slides.
TC004	Browse Themes and Load More	Test the browsing of themes and the "Load More" button on the Design page.	All themes should render, and the "Load More" button should work as expected.
TC005	Product Listing and Cards	Verify if all products and their cards are rendered on the Cakeo page.	All products and their respective details should display correctly.
TC006	Browse by Section	Validate if the "Browse by Section" feature is functional on the Cakeo page.	The feature should allow navigation between different sections successfully.
TC007	Catalogue Page	Check if the Catalogue page and its "About" section are rendered.	The Catalogue page and "About" section should be displayed without errors.
TC008	Orders Page (No Login)	Verify if the Orders page behaves correctly without a logged-in user.	The Orders page component should not render and should display a message or prompt for login.
TC009	Short First Name Validation	Verify error generation for a first name with fewer than 3 characters.	An error message should be displayed indicating the first name is too short.
TC010	Short Surname Validation	Verify error generation for a surname with fewer than 3 characters.	An error message should be displayed indicating the surname is too short.
TC011	Invalid Email Address Validation	Check if invalid email addresses are rejected during signup.	An error message should be displayed for invalid email formats.
TC012	Invalid Mobile Number Validation	Validate errors for invalid mobile numbers (e.g., short, incorrect formats).	An error message should be displayed for invalid mobile numbers.
TC013	Password Validation	Ensure an error is generated for passwords not meeting criteria (e.g., only characters or numbers).	An error message should be displayed for invalid passwords, requiring at least 8 characters with the criteria.

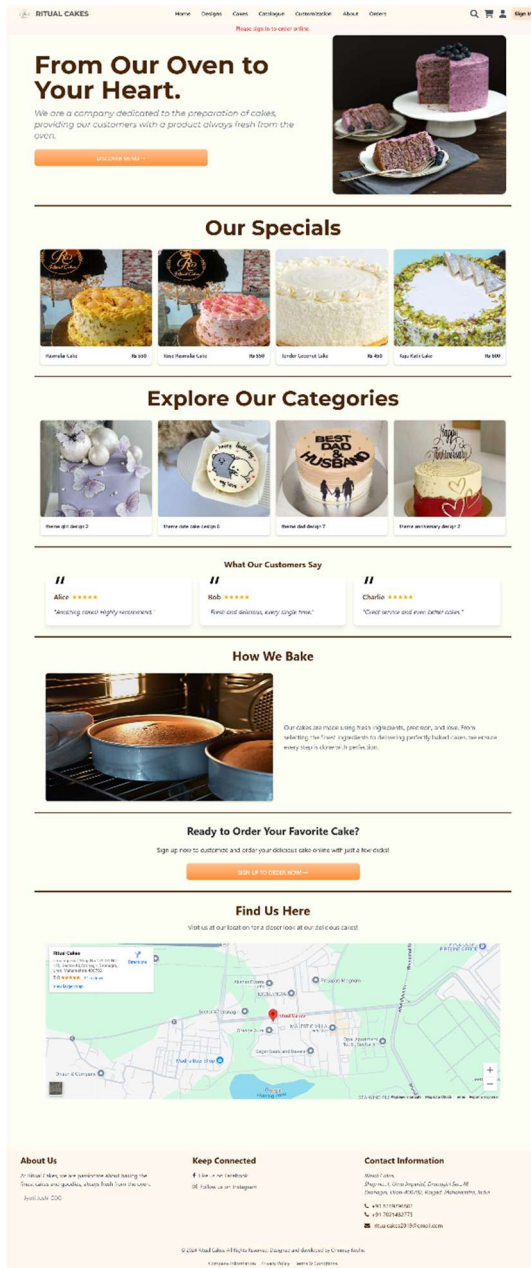
TC014	Address Field Validation	Check if errors are generated for addresses with fewer than 4 characters.	An error message should be displayed for short or invalid addresses.
TC015	Successful Signup	Verify successful signup when all fields meet criteria.	User should be registered, receive a "Welcome" email, and be redirected to the appropriate page.
TC016	Login with Invalid Credentials	Validate error generation for login attempts with invalid credentials.	An error message should be displayed indicating incorrect username or password.
TC017	Successful Login	Verify login functionality with valid credentials.	User should log in successfully and be redirected to the dashboard.
TC018	Admin Role Redirection	Check if admin users are redirected to the admin page upon login.	Admin users should be redirected to /admin and shown relevant admin privileges or warnings if necessary.
TC019	Admin Role Validation	Verify that admin privileges are granted only to specific emails and roles defined in the code.	Only users with admin roles (based on email and role in the code) should have access to admin functionalities.
TC020	Home Page Sections Rendering	Verify if all sections on the Home page are loaded correctly without login.	All sections, including the Hero, Designs, Cakes, and Footer, should render correctly.
TC021	Hero Section Loading	Test if the Hero section on the Home page is displayed properly.	Hero section should load without any issues.
TC022	Designs Page Rendering	Verify if the Designs page and its elements load correctly.	The Designs page should load fully with all expected elements visible.
TC023	Design Customization Page	Validate if the Design Customization page is loaded without errors.	The Design Customization page should render correctly and allow interaction.
TC024	Cakes Section Rendering	Test if the Cakes section on the Home page loads properly.	The Cakes section should display correctly with all items visible.
TC025	Sort and Browse Section	Verify if the Sort and Browse section on the Cakes page works properly.	The section should allow sorting and browsing, with the correct items displayed based on filters.
TC026	Product Page Functionality	Check if the Product page loads and displays product details.	Product details page should load correctly, showing all necessary product information.
TC027	Buttons Functionality	Validate if buttons across the pages (e.g., Add to Cart, Submit) work correctly.	All buttons should perform their intended functions without errors.
TC028	Add to Cart Functionality	Verify if products can be added to the cart successfully.	Products should be added to the cart, and the cart count should update.
TC029	Related Products Rendering	Test if related products are displayed correctly on the Product page.	Related products should render and display items similar to the current product.
TC030	Reviews Section Functionality	Validate if reviews are displayed and submitted correctly.	Reviews should display and be submitted/fetched without issues.

TC031	Catalogue Section Rendering	Check if the Catalogue section loads correctly.	The Catalogue section should display all items and content properly.
TC032	Customization Form Functionality	Test if the Customization form is loaded and submitted correctly.	The form should load properly, allow input, and submit successfully.
TC033	About Page Rendering	Verify if the About page is displayed correctly.	The About page should render all its content as expected.
TC034	Order Page Rendering (No Login)	Check if the Order page renders properly without login.	The Order page should render, but specific actions might be restricted until the user logs in.
TC035	Product Added to Cart	Verify if a product is added to the cart and fetched correctly in the cart page.	The product should be added to the cart, and all relevant details (quantity, price, etc.) should display.
TC036	Reviews Posting and Fetching	Validate if user reviews are posted and fetched correctly on the product page.	Reviews should display immediately after posting and fetch correctly on page reload.
TC037	User Details Editing	Check if user details can be fetched and edited successfully.	User details should display correctly, and changes should be saved and reflected immediately.
TC038	Logout Functionality	Verify if the user can log out of the system successfully.	User should be logged out and redirected to the login or home page.
TC039	Forgot Password Email	Test if the "Forgot Password" functionality sends a reset email correctly.	A reset password email should be sent to the registered email address with the correct link or OTP.
TC040	Reset Password Functionality	Validate if the user can reset their password using the link or OTP.	Password should be reset successfully, and the user should be able to log in with the new password.
TC041	Order Status Email Notification	Check if order status updates trigger an email notification.	An email should be sent to the user with the correct order status updates (e.g., confirmed, shipped).

CHAPTER 6

SCREENSHOTS AND SOURCE CODE

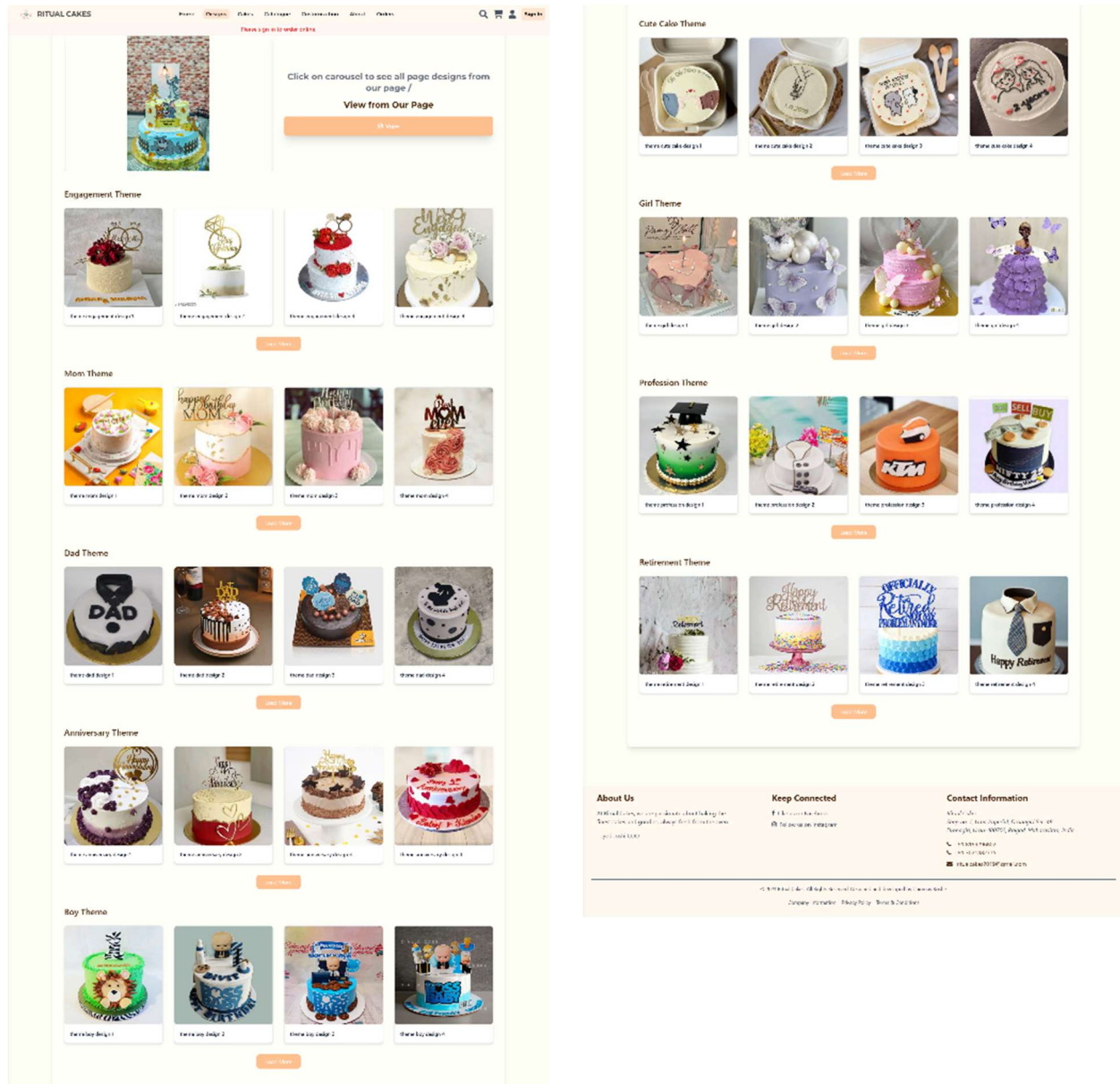
5.1 Screenshots



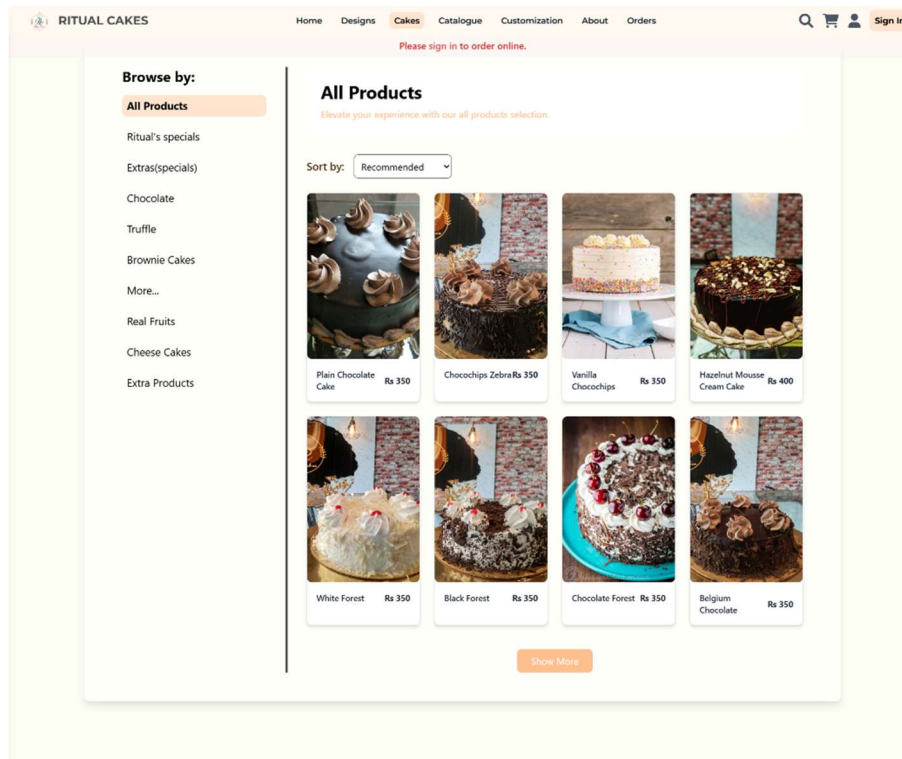
Desktop view (homepage)



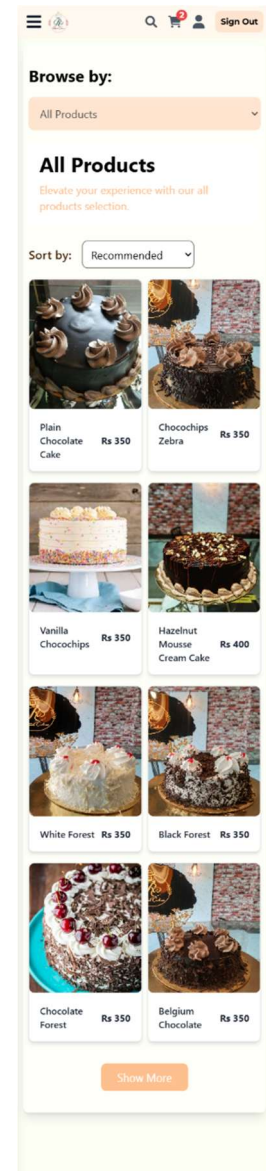
mobile view



Designs tab



Collections tab




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Rose Rasmalai Cake

Rating: 4.5 ★

Price: Rs. 550

Select Weight:

500g
1kg
1.5kg
2kg
2.5kg

Select Shape:

Round
Heart
Square

Select Quantity:

-
1
+

Add to Cart

[Description](#)
[Reviews](#)

A blend of rose flavor and traditional rasmalai.

Related Products

No related products found.

About Us

At Ritual Cakes, we are passionate about baking the finest cakes and goodies, always fresh from the oven.

Jyoti Joshi COO

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Contact Information

Ritual Cakes

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Rasmalai Cake

Rating: 4.5 ★

Price: Rs. 550

Select Weight:

500g
1kg
1.5kg
2kg
2.5kg

Select Shape:

Round
Heart
Square

Select Quantity:

-
1
+

Add to Cart

[Description](#)
[Reviews](#)

A rich and creamy cake with rasmalai flavor.

Related Products

No related products found.

Product page



Name

Phone Number

Address

Size

Cake Type

Flavor

Order Date and Time

Message on Cake

Submit Customization

Note: If the design is multi-tiered/multi-layered, please choose a weight above 1.5kg for a 2-tier cake and 2.5kg for a 3-tier cake.

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Name

Phone Number

Address

Size

Cake Type

Flavor

Order Date and Time

Message on Cake

Submit Customization

Note: If the design is multi-tiered/multi-layered, please choose a weight above 1.5kg for a 2-tier cake and 2.5kg for a 3-tier cake.

Design product page

RITUAL CAKES
Home
Designs
Cakes
Catalogue
Customization
About
Orders
Sign In

Please sign in to order online.

Customize Your Cake

Name
Full Name of orderer

Email

Phone Number
Whatsapp Number Suggested

Address
Your address

Size
Select Size

Cake Type
Select Cake Type

Flavor
Select Flavor

Order Date
Order Date must be after 2 days
dd-mm-yyyy

Message on Cake
Happy Birthday

Image URL
Enter Image URL

Special Instructions
please mention time between 10 AM TO 11 PM

Note: If the design is multi-tiered/multi-layered, please choose a weight above 1.5kg for a 2-tier cake and 2.5kg for a 3-tier cake.

Submit Customization

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Sign Out

Customize Your Cake

Name
Full Name of orderer

Email
ritualcake.admin@gmail.com

Phone Number
Whatsapp Number Suggested

Address
Your address

Size
Select Size

Cake Type
Select Cake Type

Flavor
Select Flavor

Order Date
Order Date must be after 2 days
dd-mm-yyyy

Message on Cake
Happy Birthday

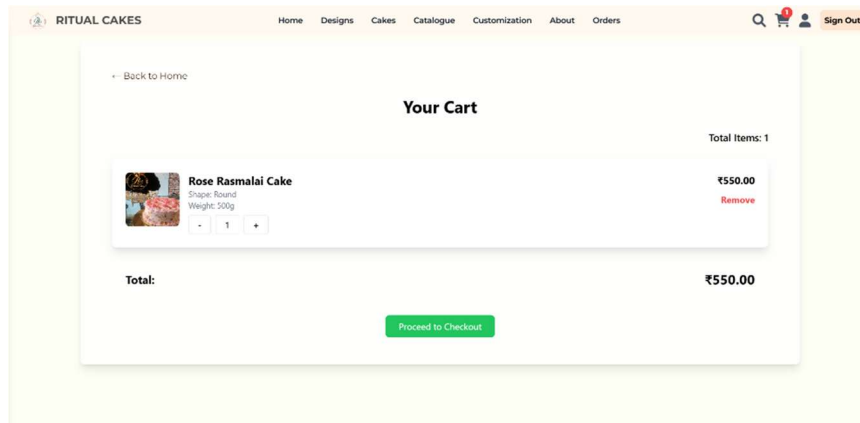
Image URL
theme_anniversary_design_2

Special Instructions
please mention time between 10 AM TO 11 PM

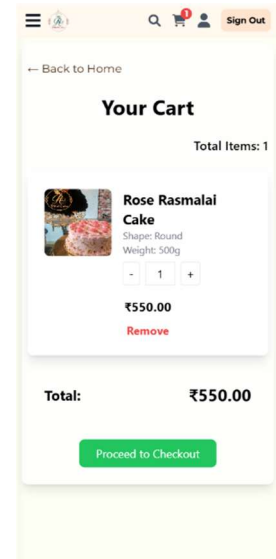
Note: If the design is multi-tiered/multi-layered, please choose a weight above 1.5kg for a 2-tier cake and 2.5kg for a 3-tier cake.

Submit Customization

Customization form



cart



The desktop view of the 'Sign In / Log In' form is centered on a light yellow background. The form has a white background and a subtle drop shadow. It includes fields for 'Email' and 'Password', a 'Forgot password?' link, a 'New Customer? Sign Up' link, an orange 'Sign In' button, and a 'Cancel' link. The website header is visible at the top, showing the 'RITUAL CAKES' logo and navigation links: Home, Designs, Cakes, Catalogue, Customization, About, and Orders. A red message 'Please sign in to order online.' is displayed below the header.

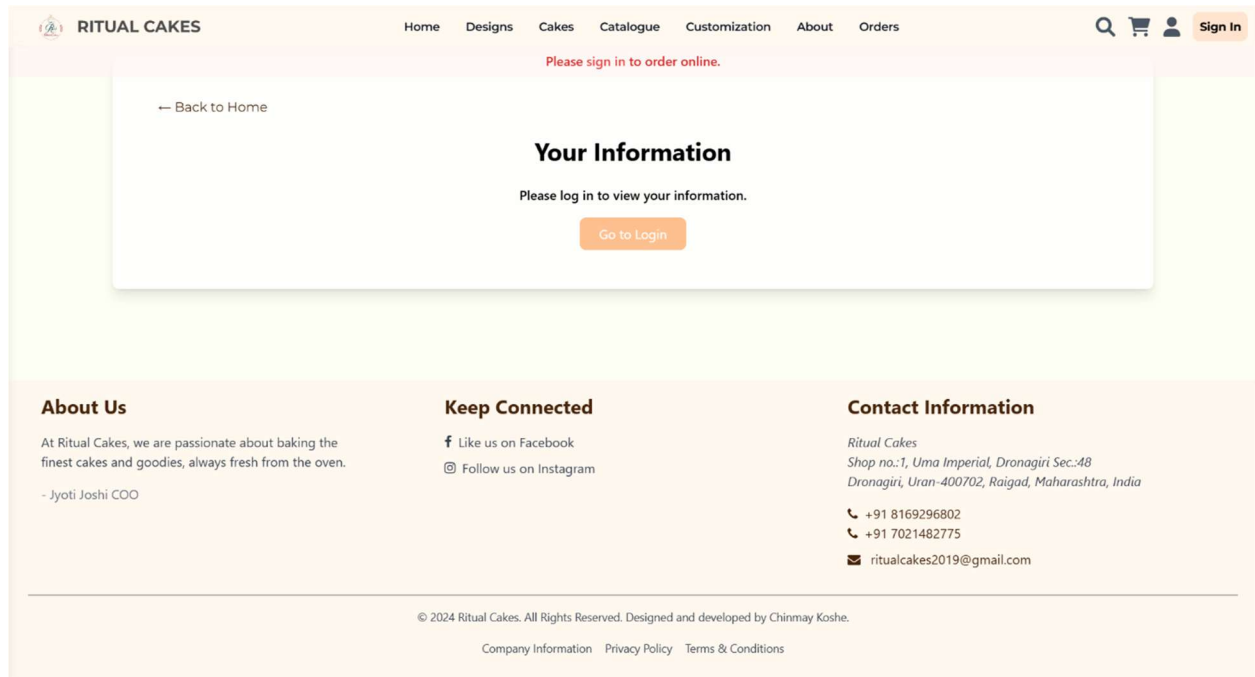
Sign in / Login

The mobile view of the 'Sign In / Log In' form is shown on a narrow screen. The form is centered and contains the same fields and links as the desktop version: 'Email', 'Password', 'Forgot password?', 'New Customer? Sign Up', an orange 'Sign In' button, and a 'Cancel' link. The mobile header is visible at the top, including the 'RITUAL CAKES' logo, navigation links, and a red message 'Please sign in to order online.'

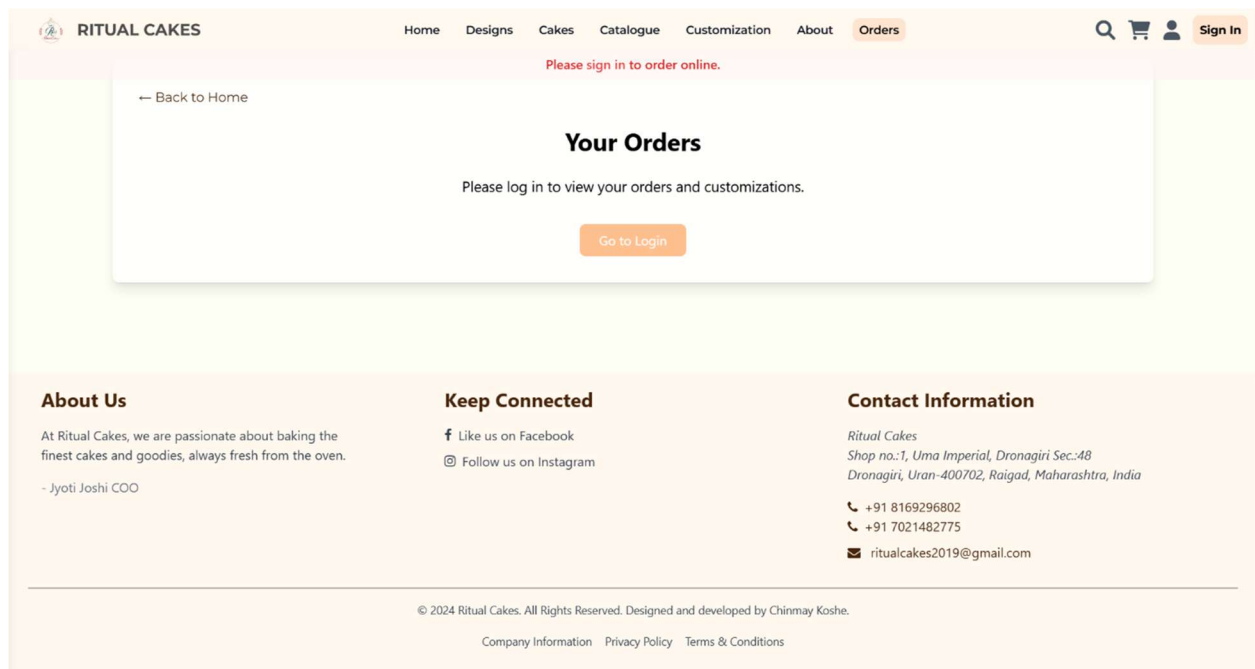
The desktop view of the 'Sign Up' form is centered on a light yellow background. The form has a white background and a subtle drop shadow. It includes fields for 'Name' (with a 'Surname' sub-label), 'Email', 'Mobile Number', 'Date of Birth' (with a date picker icon), 'Address', and 'Create Password' (with a toggle icon). There are orange 'Sign Up' and 'Reset' buttons, and a link 'Already a customer? Sign In' at the bottom. The website header is visible at the top, showing the 'RITUAL CAKES' logo and navigation links: Home, Designs, Cakes, Catalogue, Customization, About, and Orders. A red message 'Please sign in to order online.' is displayed below the header.

Sign up

The mobile view of the 'Sign Up' form is shown on a narrow screen. The form is centered and contains the same fields and links as the desktop version: 'Name', 'Surname', 'Email', 'Mobile Number', 'Date of Birth', 'Address', 'Create Password', orange 'Sign Up' and 'Reset' buttons, and a link 'Already a customer? Sign In'. The mobile header is visible at the top, including the 'RITUAL CAKES' logo, navigation links, and a red message 'Please sign in to order online.'



Profile



orders

 **RITUAL CAKES**

HomeDesignsCakesCatalogueCustomizationAboutOrders

SearchCartUser Sign Out

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Your Information

Name: Chinmay Koshe

Email: chinmaykoshe2004@gmail.com

Mobile: 9773137485

Date of Birth: 27/09/2004

Address: Omkar apt beside taramati raghunath bhoir apt Jari Mari mandir kotnaka near Axis Bank

Edit

Check Your Orders Here

About Us

At Ritual Cakes, we are passionate about baking the finest cakes and goodies, always fresh from the oven.

- Jyoti Joshi COO

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[Sign In](#)

Please sign in to order online.

About Us

Custom Bakes

At Ritual Cakes in Navi Mumbai, we offer a variety of freshly baked treats for all your special occasions. From sponge to chocolate cakes, we customize each cake to your preference...

For inquiries or to place an order, call us at [+91 7021482775](tel:+917021482775) or DM us on Instagram or Facebook.





Chocolates

In addition to our custom cakes, we offer a selection of mouth-watering chocolates like wine chocolates, stuffed chocolates, and cotton candy-stuffed chocolates...

Visit us to indulge in our delightful range of cakes and chocolates. Everyone loves cakes and chocolates, don't they?

Customer Reviews

//

John Doe ★★★★★

"Amazing cakes! Highly recommend!"

//

Jane Smith ★★★★★

"The chocolates were incredible!"

//

Emily darkcustombg1 ★★★★★

"Delicious cupcakes, will order again!"

About Us

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- Jyoti Joshi COO

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Forgot Password

Email

Send Reset Link

Reset Password

New Password



Confirm Password



Reset Password


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[← Back to Cart](#)

Checkout

Order Summary



Rose Rasmalai Cake
 Shape: Round
 Weight: 500g kg
 Quantity: 1
₹550.00

Customer Information

Your Name

Delivery Address

Message on Cake

Order Date
Order Date must be after 2 days

Select Delivery Time

Payment Method

☒ Cash on Delivery
 ☐ Online Payment

[Place Order](#)

About Us

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Plain Chocolate Cake

Rating: 4.5 ★

Price: Rs. 350

Select Weight:

500g
1kg
1.5kg
2kg
2.5kg
3kg
3.5kg
4kg

Select Shape:

Round
Heart
Square

Select Quantity:

-
1
+

Add to Cart

Description
Reviews

Customer Reviews:

Chinmay

"Awesome cakes"

Submit Review

Related Products



Chocochips Zebra

Rs 350



Vanilla Chocochips

Rs 350



Alpento Cake

Rs 350

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Your Orders



Rose Rasmalai Cake

Pending

Cancel Order

Shape: Round
Weight: 500
Quantity: 1
Amount: \$ 550

All Customizations

You have no customizations yet.

About Us

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- Jyoti Joshi COO

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Plain Chocolate Cake

Rating: 4.5 ★

Price: Rs. 350

Select Weight:

500g
1kg
1.5kg
2kg
2.5kg
3kg
3.5kg
4kg

Select Shape:

Round
Heart
Square

Select Quantity:

-
1
+

Add to Cart

Description
Reviews

Customer Reviews:

Chinmay

"Awesome cakes"

Submit Review

Related Products



Chocochips Zebra

Rs 350



Vanilla Chocochips

Rs 350



Alperito Cake

Rs 350

About Us

At Ritual Cakes, we are passionate about baking the finest cakes and goodies, always fresh from the oven.

- Jyoti Joshi COO

Keep Connected

 Like us on Facebook
 Follow us on Instagram

Contact Information

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[Terms & Conditions](#)

1 RITUAL CAKES

DASHBOARD

CUSTOMERS

ORDERS TOTAL

CUSTOMIZED ORDERS

IN STORE COLLECTION

AVAILABLE CAKES

AVAILABLE CUSTOMIZATIONS

STORE

REVIEW SECTION

VIEW STORE

HELP CENTRE

Dashboard

Welcome to the admin dashboard!

Most Sold Cake Today

Rose Rasmalai Cake

Best Sold Cake All Time

Chocochips Zebra

Total Collection Today

\$550.00

Orders

Today: 1

All Time: 42

1 RITUAL CAKES

DASHBOARD

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HELP CENTRE

Orders Panel

Search by Order ID, Email, Filter by Status: 00-9999-9999 Min Amount: Max Amount: ☒ Delete Column Show/Hide 62%

Clear All Filters

Order ID	Customer Email	Item Name	Shape	Quantity	Price	Weight	Message	Order Date	Order Time	Total	Status	Actions
679636296484180616c7	aryankul20@gmail.com	Kulat Cake	Round	1	€100	300	Happy Birthday <Chinmay>	9/12/2023	18:00	€100	Completed	Completed
679a788e7c5d5a1e87822	chinmaykech2004@gmail.com	Rose Rasmalai Cake	Round	1	€550	300	Happy Birthday Chinmay	8/11/2023	17:00	€550	Pending	Pending
679a2c17a710b8b5ef9e0	aryankul20@gmail.com	Rasmalai Cake	Round	2	€350	300	Happy Birthday	2/1/2023	18:00	€1100	Accepted	Accepted
67927072318d3d78a42404	svinshipati0@gmail.com	Black Forest	Square	1	€2000	4	Happy Birthday <name>	1/25/2023	17:00	€3800	Completed	Completed
67927072318d3d78a42404	svinshipati0@gmail.com	Rose Rasmalai Cake	Square	1	€4000	4	Happy Birthday <name>	1/25/2023	17:00	€5800	Completed	Completed
67927072318d3d78a42404	svinshipati0@gmail.com	Black Forest	Square	1	€2000	4	Happy Birthday <name>	1/25/2023	17:00	€3800	Cancelled	Cancelled
67927072318d3d78a42404	svinshipati0@gmail.com	Rose Rasmalai Cake	Square	1	€4000	4	Happy Birthday <name>	1/25/2023	17:00	€5800	Cancelled	Cancelled
67915a78810131641a2b657	jhupsh5707@gmail.com	Truffle Cake	Round	1	€400	500	Happy Birthday Jay	1/24/2023	17:00	€400	Completed	Completed
67915a78810131641a2b657	jhupsh5707@gmail.com	Truffle Cake	Round	1	€400	500	Happy Birthday <name>	1/24/2023	17:00	€400	Accepted	Accepted
67915a78810131641a2b657	jhupsh5707@gmail.com	Rasmalai Cake	Round	2	€550	500	Happy Birthday <name>	1/24/2023	17:00	€1100	Accepted	Accepted
679020950724463a712482	svinshipati0@gmail.com	Mango Cake	Round	1	€300	300	Happy Birthday	1/28/2023	17:00	€1000	Completed	Completed
679020950724463a712482	svinshipati0@gmail.com	Plain Chocolate Cake	Round	1	€150	300	Happy Birthday	1/28/2023	17:00	€1000	Completed	Completed
679020950724463a712482	svinshipati0@gmail.com	Vanilla Chocochips	Round	1	€150	300	Happy Birthday	1/28/2023	17:00	€1000	Completed	Completed
678932a2a0094040481027	kacharepadmaja10@gmail.com	Chocolate Forest	Round	1	€1000	1.5	Happy Birthday <name>	1/22/2023	17:00	€1050	Cancelled	Cancelled
678932a2a0094040481027	kacharepadmaja10@gmail.com	Chocochips Zebra	Round	1	€520	300	Happy Birthday <padmaja>	1/22/2023	17:00	€250	Accepted	Accepted
67880911c3a3e7a778a5e4	satishkale22@gmail.com	Plain Chocolate Cake	Round	1	€550	500	hello	1/21/2023	17:00	€550	Pending	Pending
67855623c18c7ef51c9e62	rohit@gmail.com	Plain Chocolate Cake	Heart	1	€200	300	Happy Birthday <name>	1/21/2023	17:00	€1450	Pending	Pending
67855623c18c7ef51c9e62	rohit@gmail.com	Rasmalai Cake	Round	2	€350	300	Happy Birthday <name>	1/21/2023	17:00	€1450	Pending	Pending
67803822a259381a723997	svinshipati0@gmail.com	Chocochips Zebra	Round	4	€250	300	Happy Birthday <name>	1/21/2023	17:00	€1800	Accepted	Accepted
67803822a259381a723997	svinshipati0@gmail.com	Hazelnut Mousse Cream Cake	Round	1	€400	300	Happy Birthday <name>	1/21/2023	17:00	€1800	Accepted	Accepted
678a10d3d3d8b8d4e15101b	chinmaykech2004@gmail.com	Hazelnut Mousse Cream Cake	Round	1	€400	500g	Ordered from store	1/16/2023	1:05:40 AM	€400	Accepted	Accepted

1 RITUAL CAKES

DASHBOARD

CUSTOMERS

ORDERS TOTAL

CUSTOMIZED ORDERS

IN STORE COLLECTION

AVAILABLE CAKES

AVAILABLE CUSTOMIZATIONS

STORE

REVIEW SECTION

VIEW STORE

HELP CENTRE

Customers List

Dashboard / Customers

Search by name, email, id

Name	Surname	Email	Phone	Address	Cart Products	Actions
Chinmay	Kochhe	john.doe@example.co	1231567890	Urban address	Rose Rasmalai Cake Tender Coconut Cake Plain Chocolate Cake Chocochips Zebra	Delete
Shirish	Patil	shirishpatil@gmail.com	0892091107	Dronagiri sec: 42	Rasmalai Cake Rose Rasmalai Cake	Delete
Tarunay Chauhan	d111	rohit@gmail.com	1111111111	11111111	Plain Chocolate Cake Rasmalai Cake	Delete
Sahil	Thakre	sahil.thakre22@gmail.com	9096745933	Urban	Plain Chocolate Cake	Delete
Aryan	Koli	aryankul20@gmail.com	9853111471	Mora	No products	Delete
Padmaja	Kachare	kacharepadmaja10@gmail.com	8591427598	Ahmad apt., mora road, uran	Chocolate Forest	Delete
Jitendra	Joish	jhupsh5707@gmail.com	8169296802	Mumbai	Truffle Cake Plain Chocolate Cake	Delete
Shirish	Patil	shirishpatil@gmail.com	0892091107	Dronagiri	Black Forest Rose Rasmalai Cake	Delete
Chinmay	K012	john.doe@example.co	1234567890	Urban	No products	Delete
Chinmay	Kochhe	chinmaykech2004@gmail.com	1234567890	Urban	Rose Rasmalai Cake Hazelnut Mousse Cream Cake Kaju Katli Cake	Delete
Shirish	Patil	shirishpatil22@gmail.com	8602091107	Dronagiri	No products	Delete
Shi	Patil	shirishpatil22@gmail.com	8602091107	Dronagiri	Chocochips Zebra	Delete
Hlex	Coder	notfocoder@gmail.com	4545454544	Urapghs	Plain Chocolate Cake Rasmalai Cake Rose Rasmalai Cake	Delete
Chinmay	Kochhe	chinmaykech2004@gmail.com	9773121745	Omkar apt beside taranpath ragnurath bhor apt Jan Man manoir kotnaka near Axis Bank	No products	Delete

RITUAL CAKES	Store					
	Search for cakes...					
	Name	Weight	Selected Price	Shape	Contact	Action
	Plain Chocolate Cake	500g	350	Round	Enter mobile number (optional)	Place Order
	Chocochips Zebra	500g	350	Round	Enter mobile number (optional)	Place Order
	Vanilla Chocochips	500g	350	Round	Enter mobile number (optional)	Place Order
	Hazelnut Mousse Cream Cake	500g	400	Round	Enter mobile number (optional)	Place Order
	White Forest	500g	350	Round	Enter mobile number (optional)	Place Order
	Black Forest	500g	350	Round	Enter mobile number (optional)	Place Order
	Chocolate Forest	500g	350	Round	Enter mobile number (optional)	Place Order
	Belgium Chocolate	500g	350	Round	Enter mobile number (optional)	Place Order
	Coffee Cake	500g	350	Round	Enter mobile number (optional)	Place Order
	Roasted Almond	500g	400	Round	Enter mobile number (optional)	Place Order
	Dairymilk Cake	500g	450	Round	Enter mobile number (optional)	Place Order
	Dryfruits Cake	500g	500	Round	Enter mobile number (optional)	Place Order
	Hazelnut Nutella Cake	500g	500	Round	Enter mobile number (optional)	Place Order
	Dairymilk Dryfruits Cake	500g	600	Round	Enter mobile number (optional)	Place Order
	KitKat Cake	500g	700	Round	Enter mobile number (optional)	Place Order
	Alpento Cake	500g	350	Round	Enter mobile number (optional)	Place Order
	Casatta Cake	500g	400	Round	Enter mobile number (optional)	Place Order
	Celebration Cake	500g	400	Round	Enter mobile number (optional)	Place Order
	Truffle Cake	500g	400	Round	Enter mobile number (optional)	Place Order
	Chocochips Truffle Cake	500g	450	Round	Enter mobile number (optional)	Place Order
	White Truffle Cake	500g	400	Round	Enter mobile number (optional)	Place Order
	White Chocochips Truffle Cake	500g	450	Round	Enter mobile number (optional)	Place Order
	Roasted Almond Truffle Cake	500g	500	Round	Enter mobile number (optional)	Place Order
	Brownie Cake	500g	500	Round	Enter mobile number (optional)	Place Order
	Walnut Brownie Cake	500g	600	Round	Enter mobile number (optional)	Place Order
	Cashew Nut Brownie Cake	500g	600	Round	Enter mobile number (optional)	Place Order
	Vanilla Cake	500g	300	Round	Enter mobile number (optional)	Place Order
	Mango Cake	500g	300	Round	Enter mobile number (optional)	Place Order
	Strawberry Cake	500g	300	Round	Enter mobile number (optional)	Place Order
	Blueberry Cake	500g	300	Round	Enter mobile number (optional)	Place Order
	Litchi Cake	500g	300	Round	Enter mobile number (optional)	Place Order
	Bubble Gum Cake	500g	300	Round	Enter mobile number (optional)	Place Order
	Opera Cake	500g	300	Round	Enter mobile number (optional)	Place Order
	Berry Berry Cake	500g	300	Round	Enter mobile number (optional)	Place Order
	Butterscotch Cake	500g	350	Round	Enter mobile number (optional)	Place Order

RITUAL CAKES

DASHBOARD
CUSTOMERS
ORDERS TOTAL
CUSTOMIZED ORDERS
IN STORE COLLECTION

AVAILABLE CUSTOMIZATIONS

STORE

REVIEW SECTION

VIEW STORE
HELP CENTRE

Available Cakes

Search cakes...

Cake Name	Order ID	Image
Pump Chocolate Cake	CHOCOLATECAKE1	
Chocolatey Zebra	CHOCOLATECAKE2	
Vanilla Cheesecake	CHOCOLATECAKE3	
Mixednut Moose Cream Cake	CHOCOLATECAKE4	
White Forest	CHOCOLATECAKE5	
Black Forest	CHOCOLATECAKE6	
Chocolate Truffle	CHOCOLATECAKE7	

Cake Name	Order ID	Image
Opera Cake	RITUALS07	
Berry Berry Cake	RITUALS08	
Sunrise Strich Cake	RITUALS09	
Berry Cust Cake	RITUALS10	
Tender Coconut Cake	RITUALS1	
Angels Kiss	RITUALS2	
Rose Petalade Cake	RITUALS3	
Kaffir Lime Cake	RITUALS4	
Rum Baba Cake	RITUALS5	

RITUAL CAKES

DASHBOARD
CUSTOMERS
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IN STORE COLLECTION

AVAILABLE CUSTOMIZATIONS

STORE

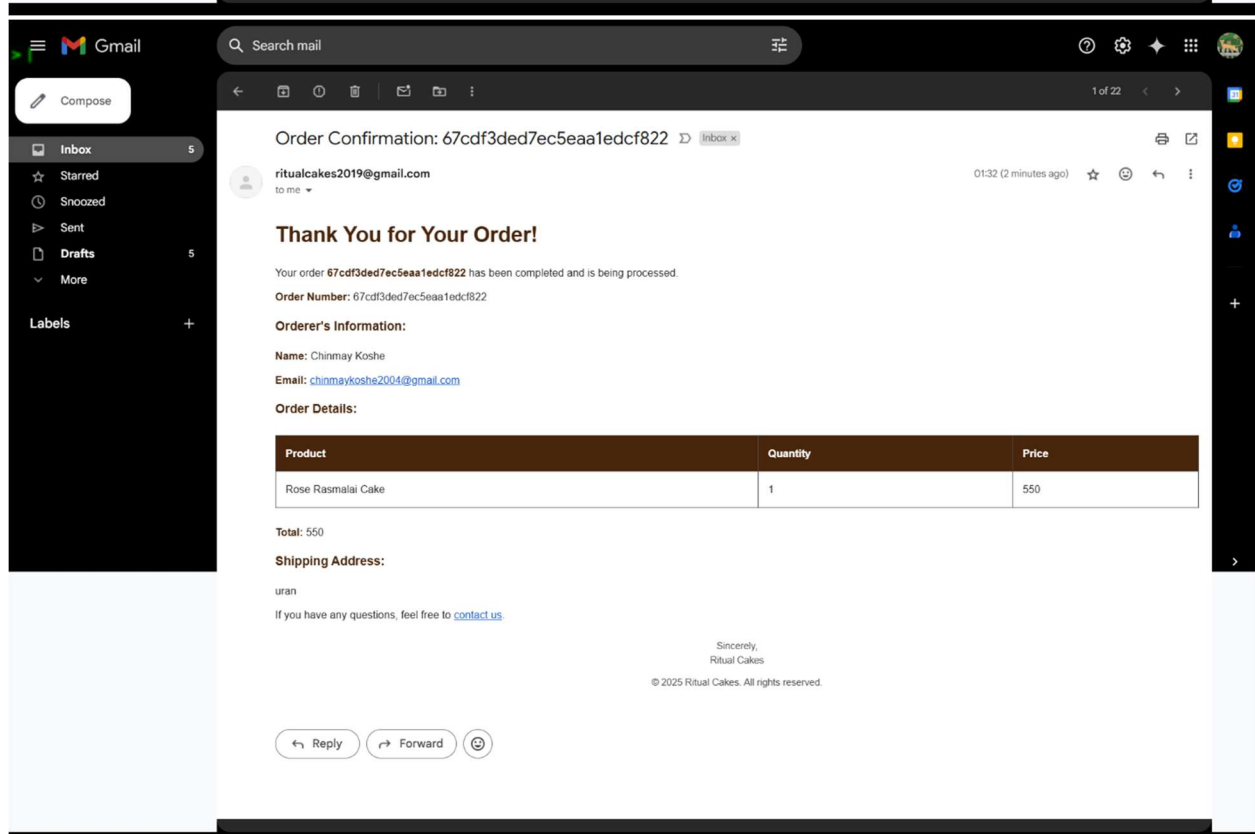
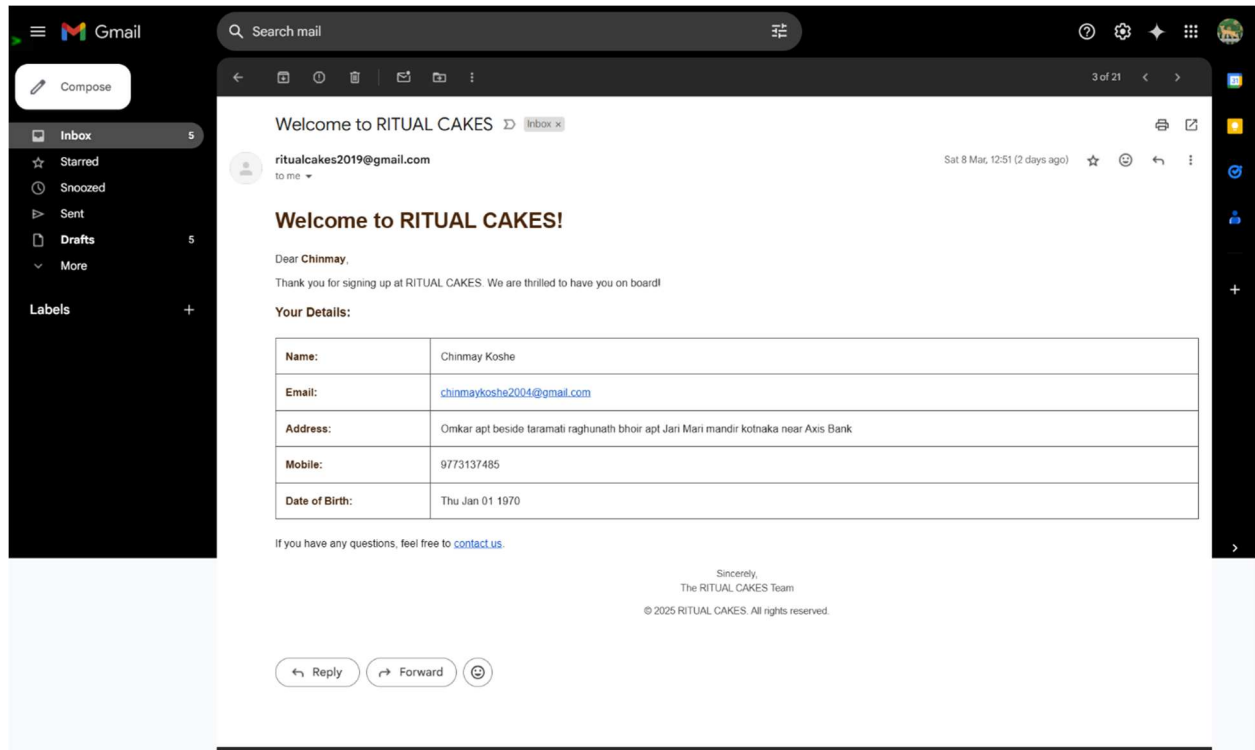
REVIEW SECTION

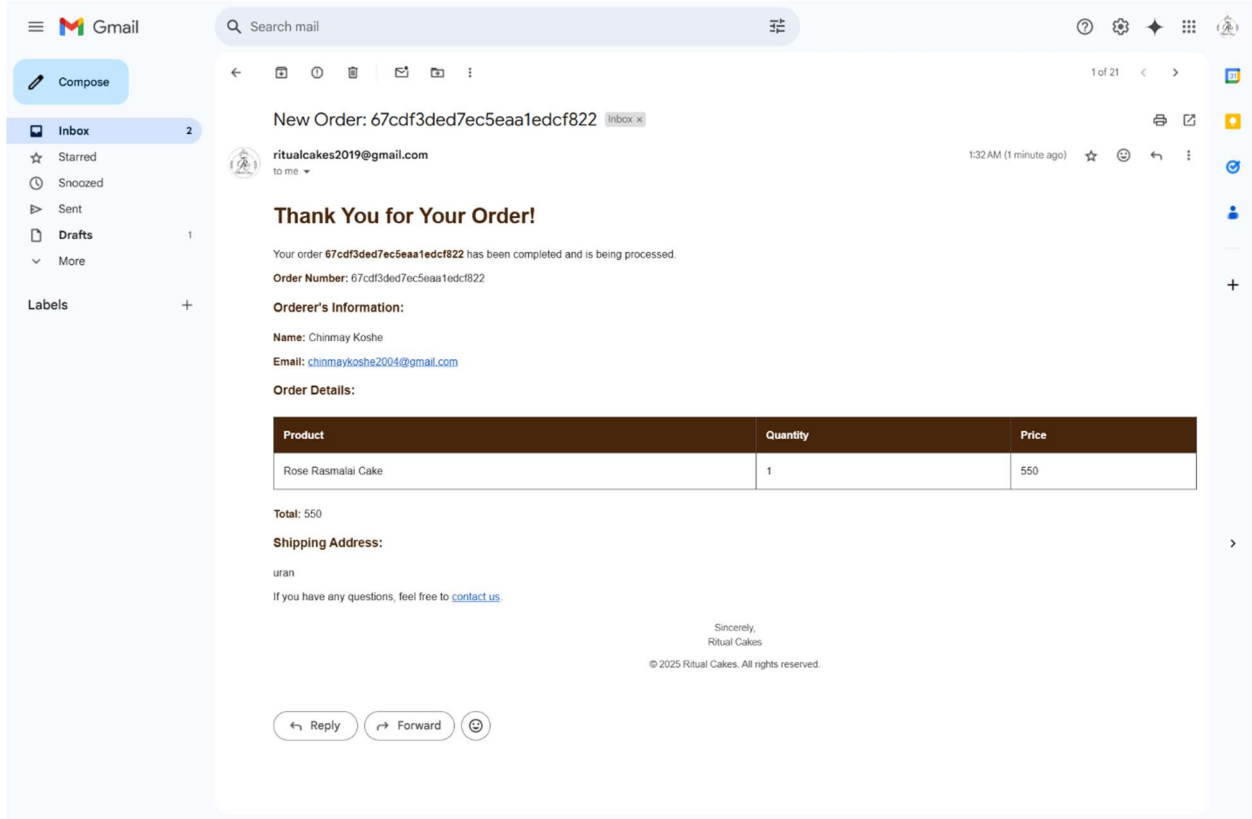
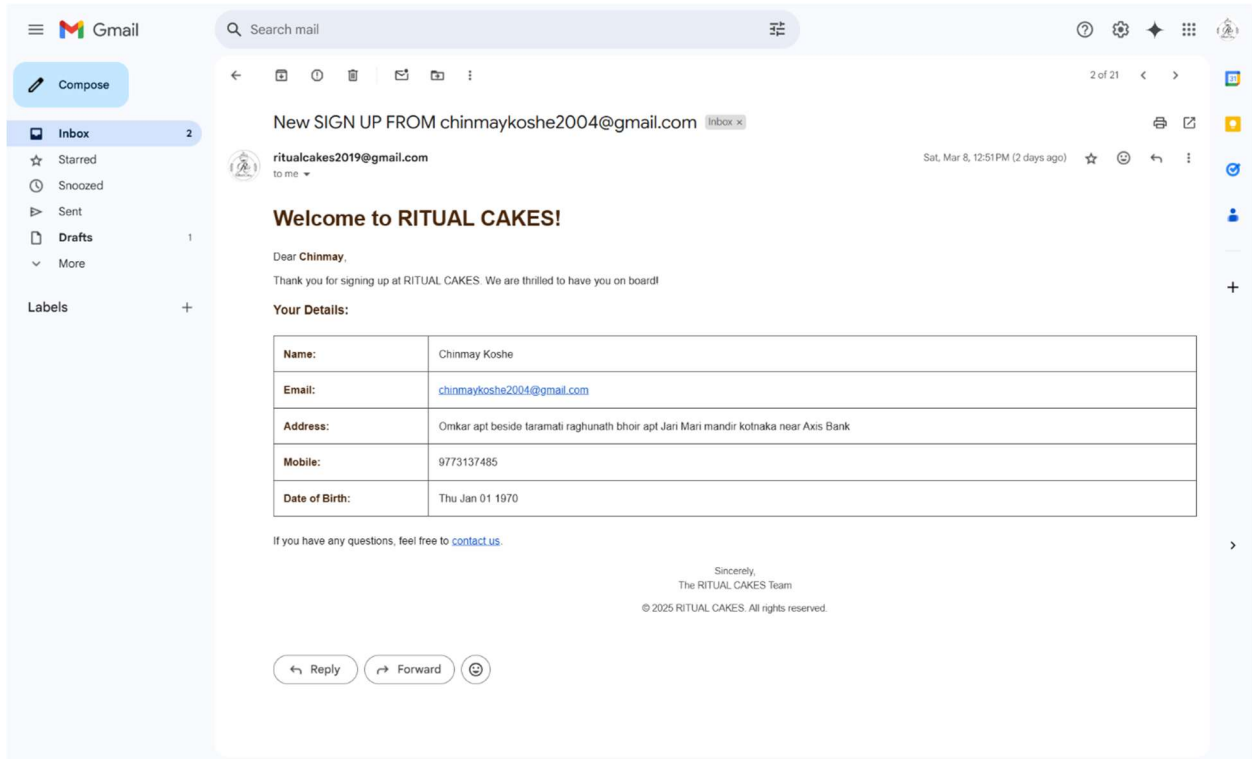
VIEW STORE
HELP CENTRE

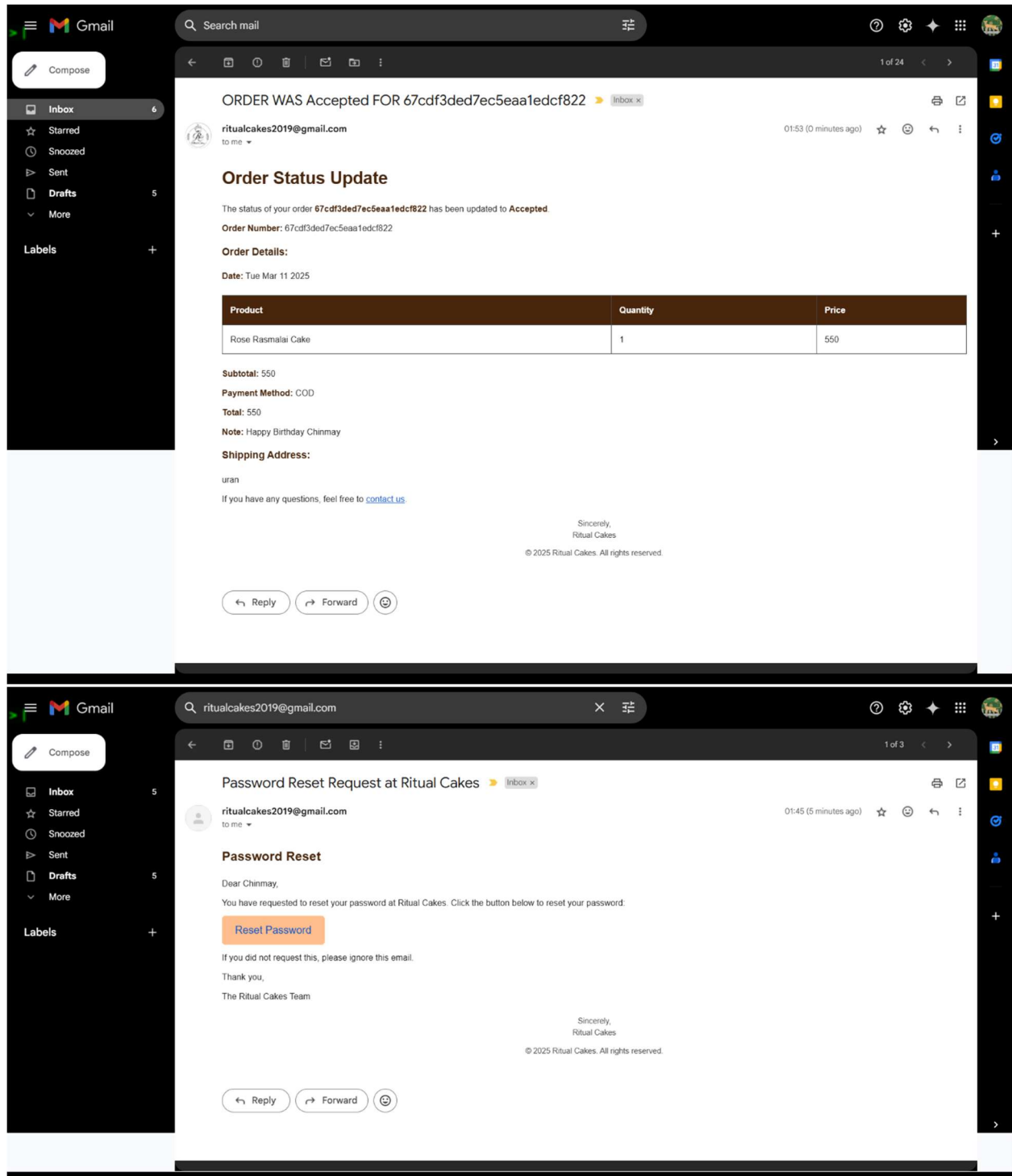
All Reviews

Search reviews...

Order ID	Reviews
CHOCOLATECAKE1	shrishtipatil85@gmail.com: Awesome cakes EditDelete
TRUFFLECAKE4	shrishtipatil85@gmail.com: Awesome cake as always 🌟🌟 EditDelete
RITUALS5	aryankoli286@gmail.com: Nice taste 🍷 EditDelete
CHOCOLATECAKE2	shrishtipatil85@gmail.com: Fresh and tasty every time worth for money 🍰 EditDelete







5.2 Source code:

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

The Cake Shop E-Commerce Website, developed using the MERN stack (MongoDB, Express.js, React, Node.js), successfully provides a user-friendly platform for online cake ordering. Key achievements include:

User Experience: An intuitive interface with easy navigation, comprehensive product listings, and customizable cake options.

Backend Functionality: Efficient user authentication, order processing, and flexible data management with MongoDB.

Security: Strong security measures, including JWT-based authentication and secure payment processing, ensuring user data protection.

Real-time Features: Enhanced transparency through real-time order status notifications.

Admin Capabilities: A robust dashboard for effective management of users and orders.

The project effectively meets the demands for a seamless online ordering experience, setting a solid foundation for future enhancements.

The Cake Shop Website project represents significant progress in online cake ordering, addressing customer needs and enhancing convenience. With planned enhancements, the platform is poised to evolve into a more versatile and scalable solution.

6.2 Significance of the System

The Cake Shop Website serves several important functions:

1. **Enhanced User Experience:** Provides a seamless shopping experience for customers with intuitive navigation & easy order customization.
2. **Efficient Management:** Streamlines administrative tasks such as user management, order updates, and order processing, allowing admins to operate more effectively.
3. **Data-Driven Insights:** Enables the generation of reports and analytics to understand sales trends and customer behaviour, supporting informed business decisions.
4. **Accessibility:** Offers online accessibility, allowing customers to browse and order cakes at their convenience, increasing potential sales.
5. **Customization:** Supports personalized cake orders, catering to individual customer preferences and enhancing customer satisfaction.

6.3 Limitations of the System

Despite its achievements, the project has some limitations:

Scalability Challenges: Currently tailored for a single shop, requiring architectural adjustments for multi-store functionality.

Lack of Advanced Analytics: The system does not provide comprehensive insights into customer behavior and sales analytics.

Lack of Payment Module: The system does not provide payment module to set the payment options for prepay.

6.4 Future Scope of the Project

The Cake Shop Website can be expanded with several enhancements:

1. **Mobile Application Development:** Create dedicated mobile apps for iOS and Android to improve accessibility.
2. **Advanced Analytics:** Develop dashboards to gain insights into customer behavior and sales performance.
3. **Multivendor Capabilities:** Transform the platform into a marketplace supporting multiple vendors.
4. **Delivery Service Integration:** Partner with logistics services to streamline order fulfillment and delivery processes.
5. **AI-Powered Recommendations:** Implement machine learning algorithms for personalized product recommendations.
6. **Augmented Reality Features:** Incorporate AR for users to visualize cakes before placing orders.
7. **Global Expansion:** Introduce multilingual support and international shipping to cater to a broader audience.

CHAPTER 7

REFERENCES

Books and Documentation:

1. **MongoDB Documentation:** Understanding database structure and querying with Mongoose.
2. **Express.js Documentation:** Guidance on routing and middleware handling.
3. **React.js Documentation:** Key resource for frontend development and state management.
4. **Node.js Documentation:** Reference for backend server construction.

Web Articles and Tutorials:

5. **MERN Stack Tutorial:** Comprehensive guide for setting up a basic MERN application.
6. **RESTful API Design:** Insights on designing API structure.

Security Resources:

7. **JWT Authentication Documentation:** Implementing secure token-based authentication.

User Interface and Design:

10. **Material-UI Documentation:** Designing responsive user interfaces.
11. **Google Maps API Documentation:** Integrating maps for store locator and delivery tracking.

Additional Resources:

12. **GitHub Repository:** Sample repositories for troubleshooting.
13. **W3Schools:** Reference for HTML, CSS, and JavaScript.
14. **NPM Packages:** Various packages like bcrypt and mongoose used in the project.